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CAPTION-ing the Situation: A Lexically-Derived Taxonomy of Psychological Situation Characteristics

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In comparison with personality taxonomic research, there has been much less advancement toward establishing an integrative taxonomy of psychological situation characteristics (similar to personality characteristics for persons). One of the main concerns has been the limited content coverage of the characteristics being used. To address this issue, we present a collection of 4 lexically based studies using the largest-to-date number of situation characteristics to identify the major dimensions of the psychological situation. These studies each implemented a unique sampling and analytic methodology—namely, a qualitative dimensional exploration; the factor analyses of 2, independent samples of large-scale in situ ratings of situations; and the use of lexical-vector representations from neural-network-based models derived from millions of sources of natural-language usage with a total of 146.7 billion words. Across these studies, a clear 7-dimensional structure emerged: Complexity, Adversity, Positive Valence, Typicality, Importance, Humor, and Negative Valence—collectively referred to as the “CAPTION” model, which parsimoniously integrates the diversity of dimensions found in the extant literature. We then introduce both full- and short-form measures of these CAPTION. Data from 2 additional diverse samples of native English speakers suggest that the measures have good psychometric properties, and are able to predict a broad range of important psychological outcomes (e.g., behaviors, affect, motivation, and need satisfaction), even when pitted against extant situation taxonomic frameworks. We conclude by discussing how the CAPTION framework may serve as a useful tool for conceptualizing and measuring a broad range of psychological situations across all areas of psychology.

Keywords: CAPTION, lexical, situation, situation characteristics, situation measure

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On the battlefield of words, where so many fights are fought in psychology, those who argue that situations play an important role in the process underlying behavior have few, if any, enemies. (Magnusson, 1984, p. 131)

There is a clear consensus in psychology that aspects of both the person and the situation are essential for understanding human behavior (e.g., Epstein & O'Brien, 1985; Fleson, 2004; Funder, 2001, 2006; Mischel, 1999; Reis, 2008). However, compared with the vast amount of knowledge accumulated on the psychology of persons, the field's progress has been substantially slower on the side of situations (e.g., Edwards & Templeton, 2005; Meyer et al., 2010; Mischel & Shoda, 1995; Reis, 2008; Ten Berge & De Raad,

1999; Yang, Read, & Miller, 2006, 2009). One particular area of deficiency in the literature is the lack of an integrative taxonomic framework of psychological situation characteristics—the major psychological dimensions of situations which differentiate one situation from another. Despite some major advances in recent years (e.g., Edwards & Templeton, 2005; Rauthmann et al., 2014; Saucier et al., 2007; Ten Berge & De Raad, 2002; Yang et al., 2006), a number of theoretical and methodological concerns persist.

One of the most frequently raised concerns is the limited content coverage of psychological situation characteristics being used, which is generally attributed to two distinct challenges. The first challenge pertains to the adoption of specific, top-down perspectives (i.e., only assessing certain types of situations such as anxiety-provoking or personality-relevant situations) that do not fully capture the content domain of situation characteristics, a point which has been continually reiterated within the literature (e.g., Cantor, Mischel, & Schwartz, 1982; Dalal et al., 2015; Edwards & Templeton, 2005; Magnusson, 1971; Ten Berge & De Raad, 1999; Yang et al., 2006, 2009). As we elaborate in the following sections, limiting the content coverage of the psychological situation characteristics under study to only those relevant to a specific focus leads to taxonomies that may systematically leave out important aspects of psychological situations, whereas the inductive (bottom-up) lexical approach has the potential to

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cover the entire domain of situation characteristics. Second, as many have noted (Edwards & Templeton, 2005; Magnusson, 1971; Ten Berge & De Raad, 1999; Yang et al., 2006, 2009), there are practical difficulties associated with data collection itself, which limits the number of situation descriptors included in the analyses, and therefore the scope, of the resulting taxonomies. This is because situations cannot rate themselves on their characteristics, as is the case for personality taxonomic research (e.g., John & Srivastava, 1999). In personality derivations, a researcher could simply gather a random sample of individuals and have them rate themselves on the items of interest. However, researchers cannot simply “recruit” a large, random sample of situations and ask “them” to rate themselves on hundreds of situation descriptors. To overcome this problem, researchers have frequently discussed the benefits of soliciting human raters’ inputs on a large, random sample of situations using hundreds of situation characteristics through a survey; however, analyzing such a large number of participants and situations using a content base of hundreds of situational characteristics has not been fully accomplished in previous research. Additionally, there are methodological alternatives to the survey method, such as qualitative and large-scale web-data explorations, but these ideas remain largely unexplored to date.

To overcome these extant challenges, our research seeks to use the lexical approach to examine the largest-to-date number of descriptors capturing psychological situation characteristics over a series of diverse studies, each employing unique sampling and analytic methodologies. Specifically, we explore these situation characteristics through the qualitative integration of trained-rater inputs; the factor analyses of two independent samples of large-scale *in situ*¹ ratings of naturally occurring situations; and the use of lexical vector representations derived from neural-network-based models calculated from corpuses of natural-language usage composed of 146.7 billion words. Across these studies, a clear seven-dimensional structure emerges: Complexity, Adversity, Positive Valence, Importance, Humor, and Negative Valence—collectively referred to as the “CAPTION” model, which fully integrates the diversity of dimensions found in the extant literature. We then introduce both full- and short-form measures of these CAPTION dimensions and provide initial validity evidence for the CAPTION measures.

Background: Taxonomic Research of Psychological Situation Characteristics

In psychology, situational characteristics are typically divided into their objective and their psychological aspects. The *objective* situation is typically defined by the characteristics of *who* is present, *what* is happening, *when* it is occurring, *where* the situation is taking place, and *which* objects are present, which are largely free from personal evaluations (Endler, 1981; Johns, 2006; Pervin, 1978; Saucier et al., 2007). For example, an objective situation could be a therapy session (*what*) with Dr. Stuart (*who*) at Psychological Services (*where*) at 9:00 a.m. on Tuesday (*when*) with a desk, chairs, and books being present (*which*). These characteristics are “objective” in the sense that, regardless of who is perceiving the situation (short of impairments in mental functioning, etc.), these characteristics “exist.” As MacIver put it, the objective situation is “the situation as it might appear to some omniscient and disinterested eye, viewing all its complex interde-

pendencies and all its endless contingencies” (MacIver, 1942, p. 295).

The *psychological* situation, on the other hand, is described with the general, evaluative characteristic(s) ascribed to the gestalt of this objective situation by an individual (Block & Block, 1981; Cantor et al., 1982; Ekehammar, 1974; Endler & Magnusson, 1976; Lewin, 1936, 1951; Murray, 1938; Rauthmann, Sherman, & Funder, 2015a; Shoda et al., 1994). Similar to personality characteristics, psychological situation characteristics can be interpreted as the perceived underlying, causal qualities which have a discernable influence on the objects and actors within the situation (Edwards & Templeton, 2005). These causal situation characteristics

might be seen in the behavior of the people in the situation (as in a ‘spirited’ discussion), on nonhuman elements of the situation (as when the effects of a ‘destructive’ battle are seen on buildings), or on the welfare, feelings, or emotions of the perceiver (as in the effects of a ‘stressful’ meeting on the perceiver’s tension level). (Edwards & Templeton, 2005, p. 707)

It is also important to note that different individuals might have different perceptions of the same objective situation. For example, one individual may perceive an objective situation (i.e., a group project meeting) to be “stressful” and “tiring,” while another may experience this same objective situation as “humorous” and “productive.” Although differences among individuals will occur, psychological situation characteristics are not simply idiosyncratic perceptions. An individual’s experience of a situation is composed of both the consensually shared perceptions of the situation—typically referred to as the consensual/canonical situation—as well as their idiosyncratic deviations from these consensual perceptions (Block & Block, 1981). For example, the situation of “skydiving” would be perceived, on average, by individuals as more “dangerous” and “scary” than the situation of “watching television” (i.e., the consensually held perceptions of these two situations). At the same time, some individuals will perceive each situation as more or less dangerous/scary as compared with others (i.e., individuals’ idiosyncratic perceptions/deviations from the consensual/canonical situation). These evaluative psychological situation characteristics are the meaning-generated, evaluative components of situations and are theorized to be the key mechanism through which objective situations influence psychological outcomes (Cantor et al., 1982; Edwards & Templeton, 2005; Ekehammar, 1974; Mischel & Shoda, 1995; Ross & Nisbett, 1991). To identify and organize these psychological characteristics of situations in a systematic, generalizable manner is the primary goal of situational taxonomic research for our field.

Major Approaches to Situation Taxonomic Research

To date, there are four major situational taxonomic approaches that have been used: environmental frameworks, frequently occurring situations, personality-relevant situations, and lexically based situation characteristics (see Table 1 for brief definitions of these approaches). Although strict delineations between the approaches are not always possible (as studies can use multiple

¹ In this article, the term “*in situ*” denotes situations that individuals experience first-hand and are psychologically immersed within, as opposed to situations that are perceived and evaluated from an external perspective.

Table 1

General Situational Taxonomic Approaches and Their Strengths and Weaknesses for Creating Taxonomies of the Psychological Situation

Approach	Definition	Examples	Can capture the full domain of psychological situation characteristics?	Can theoretically integrate the findings across the various psychological situation approaches?	Can be generalizable and replicable?
Environmental Frameworks	Assesses specific environments (work, school, basketball game, etc.) to better understand how context influences behavior.	- Basketball game, Auction Sale, etc. (Barker, 1968) - Sport, recreation, travelling, etc. (van Heck, 1984, 1989)	No, focuses on the specific environments and does not directly sample psychological situation characteristics.	No, due to focus on specific environmental frameworks.	Maybe, but more research is needed.
Frequently-Occurring Situations	Focuses on situations which are encountered frequently (anxiety producing situations, interpersonal situations, etc.) to better understand these situation types/sub-types.	- Anxiety provoking situations (Endler, Hunt, & Rosenstein, 1962) - Interpersonal situations (Kelley et al., 2003) - College students (Eckes, 1995) - Housewives (Forgas, 1976)	No, this approach assesses a subset that commonly occur for that population (e.g., college students) or specific type of situation (e.g., anxiety provoking situations) under study. Predominate focus on objective situation characteristics.	No, but can integrate the previous research related to the specific population or type of situation under study.	Somewhat—it is quite likely that the dimensions will replicate for the same population or type of situation under study, but the ability to generalize likely varies between the sub-approaches used.
Personality-Relevant Situations	Attempts to uncover situations that might manifest personality-related behaviors (conscientiousness, extraversion, etc.).	- Abridged Big Five Circumplex Personality Situations (Ten Berge & DeRaad, 2001, 2002) - California Adult Q-Sort Personality Contexts (Rauthmann et al., 2014)	No, due to the focus on only situations which might manifest personality, this approach likely excludes situations whereby personality is less likely to emerge, as well as more general, non-directly personality-relevant characteristics of situations.	No, only within personality-manifesting situations and not psychological situation characteristics, at large.	Yes, the personality manifesting dimensions should be replicable and generalizable.
Lexically-Based Psychological Situation Characteristics	Assesses the important characteristics of situations which have been encoded and maintained within the lexicon, in line with the lexical hypothesis.	- English Adjectives (Edwards & Templeton, 2005) - Chinese Idioms (Yang et al., 2006)	Yes, but a study that simultaneously assesses the full domain has yet to be conducted.	Yes, this approach can capture the full domain of psychological situation characteristics, not limited to any specific population (e.g., college students) or a priori focus (e.g., personality-relevant situations).	Yes, lexically-based taxonomies have often been theorized and found to be generalizable and replicable (John & Srivastava, 1999; Ten Berge & De Raad, 1999).

approaches), most situational taxonomic attempts can generally be considered as falling within one of these four broad approaches. Below, we discuss each of these approaches in light of their unique strengths and weaknesses based on the following overarching guidelines drawn from the literature for creating a broad taxonomy of psychological situation characteristics (Cantor et al., 1982; Dalal et al., 2015; Edwards & Templeton, 2005; Magnusson, 1971; Rauthmann et al., 2014, 2015b; Ten Berge & De Raad, 1999; Yang et al., 2006, 2009): The approach must (a) capture the full domain of the general, evaluative *psychological* situation characteristics, (b) be able to integrate the findings from diverse lines of research using the various perspectives in the extant literature, and (c) be generalizable and replicable.

Among the earliest-adopted conceptual approaches to situation classification were attempts toward a greater understanding of environmental frameworks (e.g., Barker, 1968; Price & Blashfield, 1975; van Heck, 1984, 1989). This approach might attempt to focus on what behaviors may occur within a specific environment, such as those that might occur within a workplace, a sporting event, or when traveling. As one example of this approach, Barker (1968) conducted an in-depth analysis of a basketball game, making detailed observations of the rules involved, such as the structure of the game, the roles of the observers/participants, and the dimensions of the court. Although this approach is useful in providing a deeper understanding of the structure of situations and the influence that the broader environment has upon behaviors, it does not directly attempt to identify the general, evaluative psychological situation characteristics through which individuals process their situations. Thus, although useful for providing an in-depth understanding of particular environments and the behavioral patterns within, this approach is likely not as useful in capturing the major dimensions of the psychological situation.

The frequently occurring situations approach allows researchers to identify and analyze situations that are often encountered in daily life (e.g., Battistich & Thompson, 1980; Eckes, 1995; Forgas, 1976; Kelley et al., 2003; Magnusson, 1971; Pervin, 1976). Research using this approach has typically analyzed 12 to 36 key situations commonly experienced by the population (e.g., college students, homemakers) or type of situation under study (e.g., anxiety-provoking situations, interpersonal situations), and have typically used 9 to 38 situation characteristics (e.g., situationally descriptive adjectives such as “pleasant” or “complex”) to interpret these situations (Yang et al., 2009). For example, Forgas (1976) compared the 25 most frequently encountered social situations for two populations (housewives and students). Examples of these situations included “having a short chat with the house delivery man” (housewives) and “having morning coffee with people in the department” (students). The author then interpreted these situations using 12 psychological situation characteristics (adjectives such as “complex,” “regular,” “intimate,” etc.) to help understand the relations underlying these frequently encountered social situations. The key advantage of this approach is that it can provide in-depth information on the types of situations experienced by the specific group(s) or within the particular type of situation(s) under study. However, situational frequency likely varies across populations (e.g., students, retirees), and focusing only on specific types of situations (e.g., anxiety provoking situations) limits the generalization of the derived dimensional structure, making it challenging to obtain a broad, generalizable psychological situational tax-

onomy. Additionally, these attempts often focused predominately on the objective characteristics of situations, and only assessed a limited number of psychological situation characteristics (12–36), when psychological characteristics were assessed at all. Thus, although the frequently occurring situations approach is quite useful for research focusing on a specific population or type of situation, concerns over breadth of the content coverage and the generalizability of the dimensions identified, as well as its predominant focus on the objective characteristics of situations, limit this approach’s utility in identifying a fully integrative dimensional structure of psychological situation characteristics—which is the goal of our current efforts.

Recently, taxonomic research focusing on personality-relevant situations has gained significant interest in the field (e.g., Rauthmann et al., 2014; Saucier et al., 2007; Ten Berge & De Raad, 1999, 2001, 2002). The general focus of this approach is to identify situations in which certain personality traits are expected to be behaviorally manifested over others. For example, a situation in which “minor details are important” might be more likely to elicit conscientious behavior (e.g., “detail-oriented” behaviors). The unique strength of this approach is that it attempts to directly sample contexts in which personality-relevant behaviors might be enacted, which is extremely useful for predicting personality-related outcomes. However, by focusing only on personality-relevant contexts, these attempts may not capture other important situation dimensions, such as situations in which individual differences are *less* likely to be exhibited (i.e., situations which might *restrict* the manifestation of personality), as well as the more general, less personality-relevant dimensions of situations (Dalal et al., 2015; Yang et al., 2006). Concerns have also been raised about the circularity of logic involved in defining situations by their personality-relevance (Johnson, 2015; Kreuger, 2015). These authors have questioned how much knowledge is gained when predicting behaviors, such as being “talkative” (i.e. exhibiting extraverted behavior), from situations in which “talking is expected or demanded.” Thus, although examining situations for their personality-relevant content has been undoubtedly fruitful, there are questions remaining as to whether this approach can provide the integrative, broad taxonomic structure of psychological situations that has so long been desired.

The last major approach to developing a situational taxonomy is the lexical approach (Battistich & Thompson, 1980; Eckes, 1995; Edwards & Templeton, 2005; Forgas, 1976, 1978; Pervin, 1976; Van Heck, 1984, 1989; Yang et al., 2006). The lexical approach posits that if a concept is psychologically important, it becomes encoded and maintained within a culture’s language and that these concepts hold psychologically meaningful information (e.g., Allport & Odbert, 1936; Ashton & Lee, 2005; Cattell, 1943; De Raad et al., 1998; Galton, 1884; Goldberg, 1990; Goldberg, 1982; Saucier & Goldberg, 2001). Although the lexical approach is most well-known for its successful integration of the diverse personality literature, it has also been frequently used and theorized about within the situational domain (e.g., Battistich & Thompson, 1980; Eckes, 1995; Edwards & Templeton, 2005; Forgas, 1976, 1978; Pervin, 1976; Van Heck, 1984, 1989; Yang et al., 2006). The basic argument for this approach is that language evolved as a mechanism for efficiently conveying information about aspects of our environments which are useful for the survival of the human species (e.g., where to get food, how to organize hunts, to signal

the presence of a predator, what/who/where to avoid; Corballis, 2009; Kirby, 2007; Pinker & Bloom, 1990). Therefore, the most important characteristics which differentiate one situation from another (as well as other important aspects of our environment, such as objects, persons, tastes, smells, etc.) have been encoded and maintained within the lexicon. For example, descriptors such as “nice,” “tender,” “unique,” “wonderful,” and “extravagant” can be used to describe a person (perhaps a kind parental figure with slightly lavish spending habits?), but they can also readily be used to describe a meal (a well-prepared steak?), a situation (a romantic rendezvous?), and many other important phenomena. Thus, just as they do for persons, single-term descriptors offer an efficient mechanism for perceiving, processing, and reacting to the relatively innumerable content of the objective situations that we encounter (Edwards & Templeton, 2005), as they escape “from the details and diversity of content, leaving instead the recognition . . . of generalizations and uniformities spanning situations, experiences or behaviors that insistently naïve or sensitive eyes could construe as endlessly unique” (Block & Block, 1981, p. 90).

Such a stance is useful for creating a taxonomy of psychological situation characteristics, as it implies that researchers can finally assess the entire domain of psychological situation characteristics by using the wealth of single-term psychological situation characteristics encoded within language. Indeed, with regard to personality, Ten Berge and De Raad (1999) stated this as:

Taxonomies that are constructed from a particular psychological point of view run the risk of systematically neglecting part of the personality sphere, whereas the lexical approach aiming at the complete lexicon provides a better chance to capture the whole domain. (p. 346)

As the lexical approach allows for the assessment of the full breadth of the content domain of psychological situation characteristics, it should have the ability to integrate, and to expand upon, the diversity of findings from the various situational taxonomic approaches (e.g., personality-relevant situations, emotion-provoking situations, interpersonal situations) into the broadest structure thereof. This integrative and broad content sampling also increases the likelihood that the resultant taxonomy will be generalizable and replicable across populations and subpopulations. Thus, the lexical approach to situations has the potential to satisfy all of the requirements for generating a broad, generalizable taxonomy of the major situation dimensions.

Despite its clear potential, past research utilizing a lexical approach to situational taxonomic development have largely been limited in scope. Early studies using lexically based techniques explored frequently used descriptors of situational characteristics, as provided by participants (Battistich & Thompson, 1980; Eckes, 1995; Forgas, 1976, 1978; Pervin, 1976). However, these attempts encountered the same issues facing research using frequently occurring situations, as they typically only assessed 9 to 38 situation characteristics provided by the specific population under study and, even then, were typically collected only as a means of interpreting other aspects of situations (e.g., environments or certain types of situations), not as a means to generate a generalizable taxonomy of the characteristics.

To exploit the strength of the lexical approach to situational taxonomic development, researchers have employed methods that utilize a greater number of situation characteristics. Specifically,

two studies, one using Chinese idioms (Yang et al., 2006) and another using English adjectives (Edwards & Templeton, 2005), simultaneously analyzed larger pools of situation characteristics (115–125 and 100–114, respectively). Across word lists, Edwards and Templeton (2005) found general support for the dimensions of *positivity*, *negativity*, *routineness/ease of negotiation*, and *productivity/goal achievement* in the English adjectives used. Among the Chinese idioms sampled, Yang et al. (2006) found that the clusters of *achieving one's goals*, *failing*, *having strong social bonds*, *being overwhelmed*, and *lack of vision* were typically found across samples. For both of these studies, the dimensions that emerged across samples within each paper were only relatively stable in that there were several inconsistencies in both the number and nature of the identified dimensions across the samples and word lists used (i.e., some factors did not replicate, additional factors emerged in different samples, and/or the interpretation of the dimensions changed based upon the specific characteristics and/or samples used). Aside from differences in language, the differing nature of adjectives versus idioms, and the use of factor analysis as compared to cluster analysis, the inconsistencies in the number and nature of the dimensions recovered both between and within these two studies likely stem from analyses of only small subsets (11%–13% or 100–125 characteristics) of the full lexical content domain identified by the researchers (Edwards & Templeton, 2005; Yang et al., 2006). In view of this, our goal is to build upon these prior lexically based studies to uncover the major dimensions of psychological situations by analyzing the largest sample of psychological situation characteristics used thus far across a diverse array of methodological and analytic techniques.

Study Overview

The current article presents a series of large-scale lexical studies that explore the content space of the psychological situation. These studies implement a diverse array of methodological and sampling strategies with the goal of identifying a robust, integrative framework that best captures the major similarities and differences among a large number of situation characteristics. After identifying an initial set of situational characteristics consisting of 851 adjectives, as well as a subsequently shortened list containing 535 nonredundant adjectives (Phase One), we conducted four separate studies (Phase Two), three of which each represent a unique approach toward lexical taxonomic development. We then conceptually integrated the major dimensions which emerged across these four studies (Phase Three). This integration resulted in seven major dimensions of situations: Complexity, Adversity, Positive Valence, Typicality, Importance, Humor, and Negative Valence (or the “CAPTION” framework). Lastly, based on the CAPTION framework, we developed an adjective-based measure (the CAPTION Scale or CAPTIONs) of these situational dimensions, as well as a short-form scale (Phase Four). We examined the psychometric properties of the full CAPTIONs and the short-form scale through analyses of its internal factor structure, convergent-discriminant validity with other variables (e.g., existing measures of trait-relevant contexts, situation classes, and personality), and its ability to predict important psychological outcomes (e.g., behaviors, affect, motivation, and need satisfaction).

Phase One: Preparing the Lexical Content Space of the Psychological Situation Characteristics

Historically, dictionaries have served as the starting point for the lexical derivations of personality because of their ability to capture the breadth of personality characteristics. However, a large portion of the terms found in the dictionary are rarely used, archaic, or are generally obscure (e.g., Allport & Odbert, 1936; Goldberg, 1982; Norman, 1967). Thus, we sought to identify a lexical corpus representing the full range of situational descriptors actually used in daily exchanges, as use and comprehension are integral for lexical derivations (Allport & Odbert, 1936; Goldberg, 1982; Norman, 1967; Yang, Read, & Miller, 2009). Fortunately, a database has been developed that fits this need: the SUBTLEX_{US} (Brysbaert & New, 2009). The SUBTLEX_{US} is a corpus of 51 million words compiled from the subtitles of U.S. films occurring between the years 1900 and 2007. The nature of this corpus allows for more accurate representations of the natural, spontaneous linguistic usage present in everyday interactions (Brysbaert & New, 2009). Owing to this, the SUBTLEX databases have consistently been found to more accurately represent lexicon usage and comprehension than other sources, such as the often-used Kucera and Francis (1967) database (Brysbaert & New, 2009; Dimitropoulou, Duñabeitia, Avilés, Corral, & Carreiras, 2010; Ferrand et al., 2011; van Heuven et al., 2014). The content representativeness of the SUBTLEX_{US}, in both its breadth of descriptors available (10,325 adjectives) and its direct representativeness of natural-language usage, provides a solid basis from which to begin to explore the content domain of psychological situation characteristics.

Removing Non-Situationally Relevant Adjectives

Clearly, not all of the 10,325 adjectives in the SUBTLEX_{US} represent psychological situation characteristics. Thus, the first task was to eliminate any adjectives that clearly did not describe “situations,” broadly defined. To this end, three trained research assistants and the first author each independently coded the full 10,325 adjective list using Edwards and Templeton’s (2005) framework of whether each adjective fit into either of the following sentences: “That situation was _____” or “That was a _____ situation.” Because the goal of this stage was simply to remove any adjectives that were clearly not descriptive of situations (e.g., double, khaki, vascular), any adjective identified by *any single coder* as possibly describing situations was retained for further use. Thus, 100% agreement was required to remove any adjective. This process resulted in reducing the adjective pool to 4,210 adjectives which might possibly describe situations.

Psychological Situation Adjective Identification

The next step for preparing the lexical content space was to identify the relevant characteristics of the *psychological* situation. To this end, we recruited five doctoral students in social and organizational psychology, who were trained at length on the task, its aims, and its scope. Consistent with best practices in lexical derivation studies (Goldberg, 1990), each spent several weeks independently coding the full list of 4,210 adjectives as to whether or not they represented a psychologically meaningful, evaluative dimension of situation perception, consistent with the theoretical

conceptualization of the content domain (Cantor et al., 1982; Ekehammar, 1974; Endler & Magnusson, 1976; Mischel & Shoda, 1995; Saucier et al., 2007; Shoda, Mischel, & Wright, 1994; Ten Berge & De Raad, 1999). It is common knowledge in lexical derivation research that perfect agreement between coders is nearly impossible, as strict delineations between what is and is not within the domain are unrealistic (Allport & Odbert, 1936; Ashton et al., 2004; Cattell, 1943; Goldberg, 1982; Norman, 1967). Despite this, the average pairwise agreement for the ratings of the psychological situation in the present exploration (67.4%) were slightly higher than the pairwise agreement in the lexical derivations of personality (63%; Allport & Odbert, 1936), suggesting that these ratings are likely at least as consistent as those used in the lexical derivation of personality. Consistent with the lexical derivations of personality (Allport & Odbert, 1936; Ashton et al., 2004; Cattell, 1943; Goldberg, 1982, 1990), these ratings were then finalized by the first three authors of the present article. This process resulted in a final sample of 1,133 adjectives being retained (only American English, as opposed to British, spellings of words and root words—e.g., kind as opposed to kindest, kindly—were retained).

Ensuring Reasonably Well-Understood Psychological Situation Descriptors

For the list of descriptors to be both theoretically and practically useful, it is integral that any lexically derived taxonomy only include words that are reasonably well-understood by the general population (Edwards & Templeton, 2005; Goldberg, 1982; Norman, 1967; Yang et al., 2006, 2009). To this end, we recruited 306 participants from Amazon’s Mechanical Turk (MTurk; www.MTurk.com) to identify which of the 1,133 adjectives were reasonably well-understood characteristics of the psychological situation. MTurk participants were chosen for this task as they are more representative of the general population than are typical college student samples (Barger, Behrend, Sharek, & Sinar, 2011; Buhrmester, Kwang, & Gosling, 2011). After removing seven non-native English speakers and an individual who failed the reliability checks, data from 298 respondents were retained for analysis. The participants were native English-speaking United States residents who were, on average, 38.54 years old ($SD = 13.29$), female (65%), and Caucasian (80%), with 86% of the sample having some higher education. The participants were each presented with a random subset containing 285 of the 1,133 adjectives, on which they rated their familiarity with each word presented on a dichotomous scale of “*I know the meaning*” or “*I do not know the meaning*,” similar to the lexical explorations of personality (Goldberg, 1982; Norman, 1967). Each descriptor was rated by at least 70 participants. Although there is no perfect metric for familiarity, a 96% level of comprehension was used (as an average across the familiarity ratings for each adjective), as it represents a good demarcation for what can be considered a reasonably well-understood characteristic when using samples with higher levels of education (Norman, 1967). This trimming step resulted in a final sample of 851 adjectives. Adjectives that were rated as not being very well known included such terms as *jocular*, *frenetic*, *tempestuous*, and *arduous*. The full list of the retained adjectives can be found in Appendix A; this list serves as a basis for defining the

lexical content space of the psychological situation characteristics in the current research.

Synonym Clustering of Psychological Situation Adjectives

In addition to the full, 851-adjective list, we also developed a slightly shorter list of adjectives by removing semantically redundant items from the longer list, parallel to similar attempts in lexically based personality explorations (Ashton et al., 2004; Cattell, 1943; Goldberg, 1982, 1990). To identify and eliminate the redundancies, we recruited four graduate students in Linguistics and English Literature (three doctoral-level students and one advanced Master's student). These raters were trained on the scope of the project, its aims, and the task at length. In line with best practices (Berg & Lune, 2011; Cattell, 1943; Goldberg, 1982, 1990), each spent several weeks independently coding the 851-adjective list to identify the semantically redundant situation descriptors before integrating and finalizing the findings through multiple discussion meetings held over the course of several months. Consensus from at least three of the four was required to finalize any pair of characteristics as being semantically redundant, although unanimity was typically the norm. Through this process, a total of 535 nonredundant adjectives were identified (see Appendix B).

Phase Two: Identifying the Major Dimensions of the Psychological Situation

With the content space of the psychological situation characteristics identified, we now turn our attention to the major aim of this paper: identifying the major dimensions of the psychological situation. To gain initial insights into the general structure of the psychological situation characteristics, we first conducted an integrative review of dimensions that have been identified in the extant literature. As shown in Table 2, although there is substantial divergence between studies, some consistency can also be observed. This consistency suggests that a core set of psychological situation dimensions likely exists, and that the overlapping dimensions across previous studies may be considered as particularly important. Nonetheless, because of the various challenges mentioned earlier in the current article, no research to date has captured the entire content domain of psychological situation characteristics. In view of this, in the following four studies we employed a diverse array of methodological and analytic tools to identify the core dimensions of the psychological situation—namely, the use of qualitative inputs, large-scale in situ ratings of situations, and the use of lexical vector representations from neural-network-based models.

Study 1: Qualitative Identification of Dimensions

We first sought to explore the dimensional structure of the psychological situation through the use of qualitative identification. Three of the individuals who previously identified the conceptual redundancies in the 851-adjective list were engaged for this purpose. These individuals were chosen as they had each spent the previous several months thinking, debating, and generally entrenching themselves within the content space of the psycholog-

ical situation. In addition to this expert knowledge of the content space being used, these individuals were chosen as they held no personal stake in the dimensions to be identified and had no knowledge of any previous dimensional structures which have emerged in the extant literature. Therefore, they could provide the closest approximation of an unbiased qualitatively derived dimensional structure as possible. It is also important to note that this qualitative study was conducted prior to the analysis of any data presented in the current article. Thus, it is unlikely that a subjective bias toward any specific dimensional structure, from either the subjects or the experimenters, played a role in the dimensions identified. In general, this approach toward dimension identification should help to provide a unique and important initial basis from which to integrate the more quantitatively driven approaches used in later studies.

Method

The graduate students were trained both on the nature of the task and on the general concept of dimension reduction techniques by using examples from personality psychology. In line with best practices for qualitative discussion and integration (e.g., Berg & Lune, 2011; Goldberg, 1990; Thornton et al., 2015), each expert spent several weeks independently identifying the major situational dimensions present within the 535-adjective list. Once this independent identification task was completed, the trained raters met to integrate their findings. Consistent with best practices (Berg & Lune, 2011; Merriam, 2009; Thornton et al., 2015), the ground rules of mutual respect, politeness, and the importance of minority opinions were discussed at length throughout the process, the meeting was led entirely by the raters, and an individual with training and experience in observing and recording individual and group dynamics (the first author) observed and took notes on the interactions.

Results and Discussion

Through these discussions, the trained raters identified a nine-dimensional structure which they deemed to represent the major dimensions of the psychological situation present within the 535-adjective list with reasonable degrees of both parsimony and specificity. The key results of these discussions can be found in Table 3. The first dimension identified was a positive/negative valence dimension, being represented by adjectives such as "happy" and "pleasant" as well as "unpleasant" and "uncomfortable," which generated quite a bit of discussions among raters. The general concern was about balancing the depth and the breadth of the content domain coverage. In particular, concerns surrounding the distinction of this positive/negative valence dimension from the second dimension (situations that are "harmful" and "unsafe") and the third dimension (situations that were "affectionate" and "romantic") were heavily discussed. Through these discussions, the raters eventually decided that these dimensions were unique enough to separate based upon the degree of the descriptive content (e.g., harmful and affectionate situations might represent more extreme positive and negative situations and they might have slightly different content than positive and negative situations), although this remained a point of some concern. The fourth dimension was found representing situations that occur regularly, as

Table 2
Previously Identified Dimensions of Psychological Situation Characteristics

Author(s)	Complex	Intellectual	Positive valence	Negative valence	Intimacy	Lightheartedness	Typicality	Clarity	Task-oriented	Adversity	Active	Noisy	Other
Battistich & Thompson, 1980	Involving	Intellectual Interested Competent	Pleasant Exciting Cooperative Self-accepting Satisfied Happy Energetic	Unpleasant Depressed Self-rejecting Sad Tired Self-conscious	Intimate Romantic Warm Social Polite	Lighthearted Spontaneous Reckless	Routine	Know how to behave	Task-oriented Unemotional	Tense Defensive Frustrated Anxious Uncomfortable	Active	Noisy	Assertive Submissive Independent Secure
Eckes, 1995	Complicated Emotionally involved	Interesting	Pleasant Exciting Positivity	Unpleasant Frightening Competitive Negativity	Intimate	Free Not serious Informal	Familiar	Clear Know how to behave		Tense Frustrating		Noisy	With others
Edwards & Templeton, 2005							Routine		Productivity				
Forgas, 1976	Involved		Pleasant	Unpleasant	Intimate	Informal	Regular Organized	Know how to behave		Very ill at ease	Active		
Forgas, 1978	Complex Involved Complex Involved	Interesting	Cooperative Pleasant Cooperative Pleasant	Competitive Unpleasant Competitive Unpleasant	Friendly Intimate Intimate	Flexible Informal Informal			Task-oriented	Tense Ill at ease Ill at ease		Noisy	
King & Sorrentino, 1983					Intimate	Informal		Know how to behave	Work-oriented		Active		Intentionally caused Submissive
Magnusson, 1971			Cooperative Positive	Competitive Negative	Friendly Social	Free					Active		
Moos, 1973	Involvement				Support			Order	Practical- orientation Problem- orientation				Expressiveness Autonomy
Nascimento-Schulze, 1981		Interesting	Democratic	Autocratic	Friendly Personal	Informal		Clarity	Task-oriented	Tense Intense			
Pervin, 1976													
Rauthmann et al., 2014		Interesting Intellect	Positivity	Negativity Deception	Friendly Sociality	Free			Duty	Tense Adversity			Mating
Ten Berge & De Raad, 2001						Amusement	Daily Routine			Adversity			Positioning Conduct
Ten Berge & De Raad, 2002			Pleasure	Interpersonal Conflict						Individual adversity			Social demand
Yang, Read, & Miller, 2006			Social support	Failing	Having strong social bonds				Achieving one's goals	Being overwhelmed			Lack of vision

Table 3
Subject Matter Expert–Derived Psychological Situation Dimensional Structure

Factor 1	Factor 2	Factor 2	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8	Factor 9
happy	harmful	affectionate	typical	stable	clear	scholarly	tiring	productive
pleasant	dangerous	romantic	regular	orderly	comprehensible	scientific	exhausting	important
nice	unsafe	sexual	normal	controllable	easy	strategic	restless	successful
comforting	violent	unloving*	rare*	predictable	simple	technical	active	helpful
great	destructive	unromantic*	uncommon*	unpredictable*	obvious	analytical	lively	functional
sad*	harmless*	nonsexual*	unconventional*	unstable*	incomprehensible*	unscientific*	zestful	useless*
unpleasant*			different*	disorderly*	difficult*	irrational*	mellow*	unproductive*
disagreeable*			strange*	uncontrollable*	unclear*		passive*	insignificant*
uncomfortable*				volatile*	confusing*		restful*	nonfunctional*
					uncertain*			

* Reverse-scored item.

captured by adjectives such as “typical” or “common.” The fifth dimension identified was composed of adjectives concerning the stability of the situation, typified by the adjectives “stable,” “orderly,” and “predictable.” The sixth dimension representing the level of clarity in the situation was marked by the adjectives “clear,” “comprehensible,” and “easy.” The seventh dimension represented a cognitive complexity component, marked by the items “scholarly,” “scientific,” and “analytical.” The eighth dimension captured the level of exhaustion and energy expenditure of the situation, marked by the items “tiring,” “exhaustive,” and “restless.” Lastly, the ninth dimension represented the importance and goal relevance of situations, and was marked by items such as “productive,” “important,” and “helpful.”

Just as in quantitatively based derivations (e.g., factor analytic procedures), the decision on where to stop extracting additional dimensions proved to be the most challenging task for the raters. Thus, it is not surprising that, although they were relatively satisfied that these nine dimensions represented the general content space of the psychological situation, whether or not some of the dimensions were best left distinct, or as components of larger categories, remained a concern. That said, the fact that these nine dimensions map fairly well onto the dimensions previously found in the literature (see Table 2) suggests these qualitatively derived dimensions may be accurately capturing the major dimensions of psychological situation characteristics.

Study 2: Large-Scale Ratings of Naturally Occurring Situations

In parallel to the rater-based, qualitative study (Study 1), we conducted two sets of quantitative survey-based lexical studies (Studies 2 and 3). As we discussed earlier, the major methodological concern for the generalizability of a psychological situation taxonomy is the need to drastically increase both the breadth of the situations (as reported by a large number of diverse individuals) and the number of situation characteristics being rated. With this in mind, we empirically explored the dimensional structure of the psychological situation characteristics by simultaneously assessing the largest number of situations and situational descriptors used to date in situational taxonomic efforts—specifically, situations reported by more than 1,600 individuals rated on 851 adjectives. We did so by factor-analyzing individuals’ ratings of naturally occurring situations experienced by the raters themselves (i.e., in situ).

Method

Participants and procedures. In response to the continued call for the use of larger and more diverse samples of participants to help ensure the generalizability of the dimensions identified (Cantor et al., 1982; Magnusson, 1971, 1981; Rauthmann et al., 2014; Sherman et al., 2010; Ten Berge & De Raad, 1999; Van Heck, 1989; Yang et al., 2009), we recruited 2,100 participants from MTurk, as they more accurately represent the diversities present within the population at large (Barger et al., 2011; Buhrmester et al., 2011). The participants were, on average, 39.04 years of age ($SD = 13.23$), female (67.6%), and Caucasian (79.8%). Individuals who were non-native English speakers or who failed the reliability checks were excluded from further analysis, resulting in a total sample of 1,689 respondents.

Consistent with best practices for situational taxonomic explorations (e.g., Edwards & Templeton, 2005; Rauthmann et al., 2014), participants were randomly assigned to a time, on the hour, between 8 a.m. and 8 p.m. the day before. Participants were asked to provide a description of the situation they were in at that time and were asked to rate how well each of the 851 adjectives described that situation on 5-point scale. If participants were sleeping, they were asked to rate what they did upon waking.

Analytic strategy. To identify the dimensional structure of the psychological situation, we used the factanal function in R (R Core Team, 2015) using a promax rotation. Multiple indices were used to decide how many factors to retain including parallel analysis, optimal coordinates, scree plots, and the interpretability of the dimensional structures.

Results and Discussion

The analyses indicated that between 5 and 22 factors might be conceivable based on inspection of the indices used (i.e., parallel analysis, optimal coordinates, and scree plots). However, the analyses further revealed that assuming any additional factors past a seven-factor solution resulted in factors with weak loadings (i.e., all loadings below .50), with stronger cross-loadings on the other seven factors than onto any additional factors, that accounted for little additional variance (<1%). As the seven-factor solution also revealed the most interpretable factor structure, the seven-factor solution was retained. The 10 adjectives with the highest factor loadings for each of the seven factors are presented in Table 4.

Table 4
Top Loading Adjectives for Seven-Factor Solution for Study 2

Adjective	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7
vile	.98	-.02	.02	-.11	-.12	.11	-.07
sinister	.94	.00	.02	-.05	-.21	.09	-.07
malicious	.94	-.07	.02	-.04	-.16	.09	.01
hateful	.94	-.06	.02	-.07	-.08	.09	-.05
creepy	.93	-.05	.07	-.03	-.21	.06	-.03
grotesque	.93	-.01	.00	-.07	-.12	.08	-.04
lethal	.92	.01	.00	-.03	-.19	.15	-.11
sadistic	.92	-.04	.00	-.03	-.16	.13	-.04
cruel	.92	-.03	-.01	-.03	-.09	.06	-.07
despicable	.91	.04	-.05	-.13	.02	.04	-.08
loving	.08	.89	-.03	-.49	.15	.20	.12
cherished	.02	.87	-.01	-.33	.05	.09	.11
precious	.07	.87	-.04	-.25	.06	.06	.05
special	.02	.86	-.16	-.18	.09	.10	.07
lovely	-.01	.85	-.01	-.27	-.12	.00	.06
heavenly	.20	.83	-.09	-.14	-.17	-.01	-.11
treasured	.00	.83	-.03	-.22	.02	.04	.10
tender	.19	.83	.01	-.36	.09	.12	.05
sentimental	.08	.82	-.10	-.31	.15	.03	.15
magnificent	.03	.82	-.18	-.02	-.08	-.05	.02
ordinary	.06	-.30	.73	-.13	-.16	.09	-.04
typical	.03	-.23	.70	-.14	-.07	.16	.05
common	.13	-.20	.70	-.21	-.08	.17	.01
average	.11	-.39	.69	-.13	-.12	.10	.02
standard	.05	-.19	.68	.01	-.09	.19	-.09
usual	.08	-.28	.67	-.09	-.15	.18	.01
predictable	.11	-.18	.67	-.13	.04	.15	-.02
regular	.02	-.22	.65	-.04	-.18	.21	-.05
conventional	.05	-.08	.64	-.03	-.07	.12	-.07
normal	-.10	-.09	.62	-.09	-.28	.20	-.06
analytical	-.10	-.17	.01	.87	-.02	-.07	-.16
technical	-.04	-.18	.00	.81	.00	-.03	-.13
academic	-.01	.03	-.01	.79	-.13	-.06	-.26
professional	-.03	-.09	.06	.77	-.01	.10	-.33
scholarly	.01	.07	-.05	.73	-.15	-.07	-.27
intellectual	-.23	.20	-.07	.72	-.08	-.12	-.13
educational	-.22	.10	-.05	.71	-.02	-.04	-.06
instructional	-.13	.03	.02	.71	.04	.04	-.09
scientific	.12	-.08	.01	.70	-.10	.02	-.16
informational	-.22	.09	.06	.70	-.09	-.15	-.10
exhausting	.18	-.06	-.07	.06	.71	.25	.19
tiring	.13	-.13	.02	.08	.68	.24	.17
stressful	.22	-.13	-.12	.19	.65	.17	.10
fatiguing	.18	-.09	.03	.08	.65	.18	.12
frustrating	.27	-.09	-.06	.10	.61	.04	.10
tiresome	.22	-.19	.09	.10	.60	.18	.08
hectic	.20	-.06	-.02	.05	.58	.26	.35
difficult	.17	-.12	-.16	.31	.56	.13	.12
strenuous	.30	-.06	-.05	.13	.54	.29	.12
overwhelming	.33	.02	-.12	.08	.54	.17	.12
important	.05	.26	.16	.02	.27	.69	-.13
productive	-.02	-.02	.34	.26	.17	.66	-.20
useful	-.03	-.05	.44	.18	.07	.64	-.16
beneficial	-.12	.12	.41	.05	.12	.57	-.02
effective	-.03	.04	.44	.20	.05	.56	-.12
crucial	.20	.05	.29	.09	.32	.55	-.14
vital	.13	.23	.33	.01	.21	.55	-.13
practical	.06	-.08	.60	.10	.03	.52	-.17
helpful	-.08	.12	.35	.22	.14	.50	-.09
efficient	.02	-.02	.44	.27	.01	.50	-.18
silly	.05	.33	-.01	-.29	.14	-.24	.61
goofy	.18	.32	.02	-.27	.08	-.11	.58
comical	.01	.43	.01	-.20	.06	-.20	.54
nutty	.36	.16	.03	-.16	.15	-.06	.52

Adjective	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7
rowdy	.37	.09	-.07	-.19	.18	.09	.51
funny	-.08	.49	.00	-.20	-.03	-.21	.50
humorous	-.13	.52	.00	-.15	.01	-.24	.50
wacky	.35	.16	.00	-.10	.08	-.08	.49
quirky	.06	.13	-.01	.12	.09	-.18	.48
playful	-.14	.60	-.03	-.18	.02	-.19	.47

Note. $n = 1,689$. The strongest loadings for each factor are bolded.

The first factor was composed of adjectives describing negative situations. The items loading onto this factor included “vile,” “sinister,” “malicious,” and “hateful” and was the broadest of all of the factors derived, with 290 items having loadings greater than .50. The second factor that emerged was composed of positive and intimate situations, with descriptors such as “loving,” “cherished,” “precious,” and “lovely.” Although fewer than the negative valence factor, this factor also had a large number of adjectives (117) loading above .50. The third factor was composed of normal, everyday situations and was described by words such as “ordinary,” “typical,” “common,” and “average.” A fourth factor of cognitive and general complexity was typified by words such as “analytical,” “technical,” “intellectual,” “complex,” and “intricate.” The fifth factor was composed of descriptors representing situations marked by adversity/energy expenditure, described by words such as “exhausting,” “tiring,” “stressful,” and “difficult.” The sixth factor represented important and goal-oriented situations and involved adjectives such as “important,” “productive,” “beneficial,” and “crucial.” The final factor, a humor and lightheartedness factor, marked by words such as “silly,” “goofy,” “nutty,” and “funny.” An interesting component of this *humor* factor is that it seems to be simultaneously positive and negative in valence.

Because the psychological situation is evaluative in nature, we also ran several analyses to account for the possibility of a general dimension of positive/negative evaluation confounding the results. For example, we ran an exploratory bifactor model (Jennrich & Bentler, 2012), which allows for a general factor to be controlled for while assessing the exploratory factor structure. The results were such that the general factor extracted was simply the broad first factor of negative situations, rather than a general positive/negative evaluation factor. Obviously, this caused issues for interpreting the resulting dimensions, as it forced a primary dimension to be orthogonal to all of the others. However, despite this odd orthogonality, all of the dimensions were still identified, albeit with oddly behavior factors attributable to the forced orthogonality, with the exception of the factor characterized by adversity/energy expenditure. Overall, these analyses suggest that a general factor of evaluation does not emerge and that the seven-factor solution is the most interpretable dimensional structure.

Importantly, these factors show strong similarities to the nine dimensions derived from the qualitative inputs in Study 1, as well as to those found in the extant literature. In fact, the majority of the qualitatively derived dimensions and the extant dimensions were directly replicated (*positive valence*, *negative valence*, *typicality*, *complexity/intellectuality*, *adversity/energy-expenditure*, *lightheartedness/humor*, and *importance/task-orientation*), with the remaining qualitatively derived dimensions being easily conceptualized as lower-order components of the dimensions identified in

the present investigation. Specifically, the three qualitatively derived dimensions identified by the adjectives “happy,” “harmful,” and “affectionate” can be conceptually integrated within the broader *positive valence* and *negative valence* dimensions, as the raters suggested they might. Furthermore, in addition to the qualitatively derived dimensions with the items of “typical” and “tiring,” in the two dimensions composed of adjectives such as “stable” and “clear” can be seen as subcomponents of *typicality* and *adversity*, respectively, with their content showing loadings onto both of these factors. The only clear difference between the qualitatively derived dimensions and the current study is the emergence of the *humor/lightheartedness* factor in the current study using the full 851-adjective list. However, this dimension has consistently emerged in the prior literature (lightheartedness; see Table 2). Together, these findings suggest that these seven factors might represent consistent, generalizable dimensions of the psychological situation characteristics.

Study 3: Replicability and Robustness of the Factor Structure

With a consistent factor structure beginning to emerge, we felt it was important to test whether this structure can be replicated and whether or not it is robust to changes in item content. Practically, unwanted variance might result from such a long list of items (851 descriptors) being given to participants (Ashton et al., 2004; Goldberg, 1982, 1990; Yang, Read, & Miller, 2006). It is also possible that the factors emerging might simply be due to the semantic redundancy in the item content (i.e., interchangeable synonyms), which might cause some dimensions to be overrepresented (Ashton et al., 2004; Goldberg, 1982, 1990; van Heck, 1984, 1989). Given these potential concerns, we examined the robustness of the factor structure which emerged in Study 2 by trying replicate the findings using the shorter, 535-adjective list.

Method

To continue to ensure the diverse sampling of situations, we recruited 2,000 participants from MTurk to participate in the current study who had not been part of the previous studies included in this article. Participants were, on average, 35.96 years of age ($SD = 11.71$), female (60.95%), and Caucasian (78.4%). Individuals who were non-native English speakers or whom failed the reliability checks were excluded from analysis, leaving a total sample of 1,679. The methods and data analytic plan were identical to Study 2, with the exception that participants rated the situations experienced on the nonredundant list of 535 adjectives instead of the full list of 851 adjectives.

Results and Discussion

The indices used to assess the optimal factor structure (i.e., parallel analysis, optimal coordinates, and scree plots) indicated that 5 to 18 factors might be plausible, but the six-factor solution revealed the cleanest, and most interpretable, factor structure. Rotating any factors past a six-factor solution resulted in factors with weak loadings (i.e., all loadings below .50), that showed stronger cross-loadings on the other six factors than onto any additional factors, and accounted for little additional variance

($\approx 1\%$). Therefore, the six-factor solution was retained. Ten adjectives with the highest factor loadings for each of the six-factors are presented in Table 5. These six factors are almost identical to the first six identified in Study 2: *negative valence*, *positive valence*, *typicality*, *adversity*, *complexity*, and *importance*, suggesting that these dimensions are robust to the removal of redundant items. The only difference between the two studies was the failure to replicate the *humor* factor identified in Study 2.

Given that the *humor* factor only emerged when the full list of 851 adjectives was used, we sought to explore whether its lack of replicability may be attributed to the uniqueness of the samples used in each study or to some systematic differences in the content of the 851- and 535-adjective lists. To explore this, we reanalyzed the Study 2 data using only the 535 adjectives used in Study 3. The results revealed that a nearly identical six-factor structure of *negative valence*, *positive valence*, *typicality*, *adversity*, *complexity*, and *importance* emerged in the 535 adjective portion of the Study 2 data, with strong evidence for the six-factor model as compared to models with larger numbers of factors based on the factor loadings, cross-loadings, and additional variance explained by each factor. This result, combined with components of the humor factor emerging regularly in the extant literature, suggests that the humor factor’s lack of replicability is likely due to systematic differences in the content of the 851- and 535-adjective lists rather than in the uniqueness of the study samples.

Study 3 provides further support for the *negative valence*, *positive valence*, *typicality*, *adversity*, *complexity*, and *importance* dimensions, which appear to be consistent and robust to changes in item content and derivation methods. This study also suggests that the emergence of the *humor* factor is likely influenced by changes in the conceptual content structure of the items being used. This is somewhat reminiscent of the findings in the lexical derivation of personality where the emergence of a Honesty-Humility factor was associated with changes in the item content used (e.g., Ashton et al., 2004; Lee & Ashton, 2010). Additionally, it is important to reiterate that situational characteristics related to *humor* have been repeatedly identified as a stand-alone dimension (“lightheartedness”) in prior situational taxonomic studies (see Table 2). Thus, excluding this *humor* dimension would cause the resulting taxonomy to not be able to fully integrate the major dimensions found in the extant literature. Additionally, recent research has found that the number of synonyms an adjective has is unrelated to its importance in regards to personality (Wood, 2015), further suggesting that a *humor* dimension, which was found when the full list was used, is likely an important psychological situation dimension.

Study 4: Lexical Vector Representations From Neural-Network-Based Models

Method

The first three studies in the current article showed that a relatively consistent dimensional structure of psychological situation characteristics emerged across multiple samples and derivation techniques. In the fourth study, we further examined the generalizability of this finding by exploring whether a similar structure exists in the naturalistic way in which these adjectives are used in daily language. A general confirmation of the dimensions

Table 5
Top Loading Adjectives for Six-Factor Solution for Study 3

Adjective	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
perverted	.92	-.04	-.01	-.25	-.03	.13
inhumane	.90	-.07	-.04	-.09	-.05	.10
sleazy	.90	-.06	-.01	-.17	-.01	.07
malicious	.90	-.04	-.08	-.14	-.02	.07
grotesque	.90	-.04	-.06	-.11	-.04	.12
cruel	.89	-.05	-.08	.03	-.10	.10
scandalous	.89	-.04	-.07	-.20	.04	.09
morbid	.89	-.05	-.03	-.15	-.02	.09
wicked	.88	-.01	-.06	-.10	-.01	.06
psychotic	.88	.00	-.07	-.09	-.03	.07
heartwarming	-.06	.87	.03	.00	-.28	-.01
cherished	-.01	.86	.03	.02	-.29	.05
special	-.11	.85	-.13	.08	-.12	.01
sentimental	.05	.81	-.01	.09	-.27	-.03
joyful	-.14	.80	-.03	-.19	-.17	-.09
affectionate	.06	.80	.07	.02	-.35	.07
memorable	-.15	.80	-.23	.17	.00	-.11
priceless	.02	.79	-.02	.11	-.20	.05
endearing	.00	.79	.07	.01	-.19	.01
amazing	-.04	.78	-.20	-.06	-.01	-.04
uneventful	.15	-.40	.64	-.13	-.07	.03
predictable	.06	-.19	.62	-.10	-.07	.19
typical	.02	-.24	.60	-.22	-.06	.13
unremarkable	.14	-.47	.58	-.06	.02	-.03
simple	.06	.12	.56	-.36	-.30	.06
regular	-.03	-.18	.55	-.23	-.03	.15
minor	.16	-.15	.55	-.16	-.03	-.17
simplistic	.12	.14	.55	-.19	-.28	.02
forgettable	.24	-.49	.54	-.09	.03	-.02
uncontroversial	-.11	-.07	.54	.02	.00	.04
stressful	.17	-.12	-.11	.66	.18	.12
exhausting	.13	.02	-.09	.64	.10	.20
tiring	.06	.02	.01	.63	.06	.16
difficult	.13	-.01	-.21	.61	.25	.07
strenuous	.18	.06	-.07	.59	.06	.23
unpleasant	.43	-.17	.02	.59	-.12	.13
frustrating	.22	-.15	-.02	.57	.13	-.03
uncomfortable	.45	-.06	-.03	.53	-.10	.13
inconvenient	.39	-.07	.08	.53	-.11	.11
overwhelming	.27	.09	-.15	.53	.08	.07
analytical	-.12	-.21	-.01	.05	.82	-.07
scholarly	.02	-.15	-.03	-.06	.75	.00
technical	.00	-.30	.03	.07	.72	.03
intellectual	-.21	.09	-.04	-.04	.70	-.06
professional	-.04	-.23	.07	.07	.69	.21
scientific	.12	-.14	.04	-.01	.63	.06
educational	-.21	.12	-.05	.06	.62	-.05
intelligent	-.21	.15	.04	.00	.59	.09
complex	-.04	.05	-.15	.32	.56	-.04
entrepreneurial	.18	-.04	.00	-.03	.56	.08
productive	-.05	-.07	.19	.11	.24	.67
important	-.06	.19	.12	.26	.02	.60
effective	-.08	.04	.31	.06	.20	.54
efficient	.01	-.09	.30	-.05	.25	.53
beneficial	-.16	.18	.33	.08	.03	.53
crucial	.17	-.03	.19	.31	.08	.48
proactive	-.03	.10	.21	.18	.20	.48
helpful	-.08	.17	.31	.06	.11	.47
practical	-.07	-.14	.48	-.01	.13	.47
successful	-.15	.19	.17	.00	.14	.44

Note. $n = 1,679$. The strongest loadings for each factor are bolded.

previously uncovered within this unique methodology would provide additional support for the generalizability and robustness of the dimensional structure identified. As detailed below, we analyzed the semantic similarities among the top-loading adjectives for each of the seven factors identified in Study 2 by employing lexical vector representations from neural-network-based models (Mikolov, Chen, Corrado, & Dean, 2013; Pennington, Socher, & Manning, 2014) derived from large (1.6 to 100 billion word) text corpora of natural-language usage.

Neural network-based lexical vector representations allow for the calculation of the semantic similarity among words. They were originally a byproduct of neural language modeling, where the technique was created to model word sequences. Specifically, at each (lexical) position in a sentence (the first word, second word, third word, etc.), the objective of a neural language model is to predict the upcoming (i.e., next) word given the contextual history (the words from the beginning of the sentence to the current lexical position). Initially, vector representations were considered a by-product of this modeling process, and were simply used as a computational representation for the words which were being modeled. However, because of their ability to computationally assess lexical semantics, these vectors are now used across a broad range of computational linguistics applications such as named entity recognition and modeling lexical semantic relations (Mikolov, Yih, & Zweig, 2013; Pennington et al., 2014). In the current study, we used these vector representations to calculate the semantic distances among the relevant situational characteristics by employing them using models based upon millions of samples of naturally occurring language usage in its entirety, not limited to only situationally descriptive adjectives. If a similar dimensional structure to those found in the previous three studies in this paper emerges through this unique methodological approach, then it would provide strong additional evidence for the consistency and generalizability of this dimensional representation of the psychological situation characteristics.

For calculating the semantic distances between our adjectives using their corresponding vector representations, we used neural-network-based models, which have been shown to consistently out-perform traditional distributional methods across several semantic tasks including lexical similarity, word analogy, and named entity recognition (Collobert & Weston, 2008; Mikolov et al., 2013; Pennington et al., 2014). Specifically, we used two neural-network-based models, word2vec (Mikolov, Chen, et al., 2013) and GloVe (Pennington et al., 2014) to calculate the vector representations. These models are built upon the distributional hypothesis (Harris, 1954), which states that words that are frequently used together within similar contexts (i.e., similar sentences) tend to have similar meanings. Thus, co-occurrence is used as a proxy of semantic similarity. That is, if two words co-occur within a frame of a prescribed size (e.g., within five words of one another), they are considered semantically related and their corresponding vectors are adjusted to be closer in the Euclidean space (i.e., the number representing their relatedness becomes closer to zero). It is important to note that relatedness is, in fact, different from similarity, and that this difference often leads to an important limitation of the distributional hypothesis, where related words (e.g., movie-popcorn, Finkelstein et al., 2001) are often (incorrectly) considered semantically similar, causing an increased false-positive rate when using distributional approaches. However, despite these false-

positives, true positive rates in these models are much more prevalent, and the word2vec and GloVe models, in particular, have demonstrated good performance across multiple semantic benchmarking tests including semantic similarity (Miller & Charles, 1991; Rubenstein & Goodenough, 1965) and word analogy (Mikolov et al., 2013; Turney et al., 2003).

Each of the vector representations were calculated using large, naturally occurring text corpora composed of billions of words. The word2vec vectors were calculated on a proprietary 100-billion-word subset of the Google News dataset. In addition to the word2vec vectors, three sets of GloVe vectors are used. The sets of GloVe vectors were calculated on a 42-billion-word Common Crawl dataset (www.commoncrawl.org), the English Gigaword Corpus (4 billion words, Graff et al., 2003), and a 2014 English Wikipedia dump (www.dumps.wikimedia.org/enwiki) of over 1.6 billion words. Together, these corpora represent the naturally occurring language usage of a combined 146.7 billion words compiled from millions of sources, providing a large degree of generalizability to the resultant vectors representations.

For our study, we calculated the vector representations of the top 10 loading adjectives for each of the seven factors identified in Study 2 (the adjectives presented in Table 4). These factor markers were chosen as they were drawn from the most comprehensive adjective list (851 adjectives) and also allow for an additional confirmation of the replicability of the humor factor within a different analytic technique. Although an analysis using the full list would have been preferable, due because of both the inherent challenges in using the distributional hypothesis (e.g., increased false-positive rates) and the fact that the resulting vectors are compiled into an $n \times n$ matrix of distances within Euclidean space, rather than raw scores in an observations by variables matrix (which would be amenable to other procedures, such as factor analysis), we use the 70-item list from Study 2 and employ cluster analysis to attempt to confirm that a similar conceptual structure occurs within the semantic vector space. Overall, this technique provides a unique basis from which to attempt to confirm the dimensional structure uncovered across the previous three studies.

After normalizing the vectors, we calculated the final vector representation of the adjectives by averaging across the four vector models to remove any idiosyncrasies which may be present within a specific vector calculation technique. These vectors were then subjected to hierarchical cluster analysis using the `hclust` function in R (R Core Team, 2015) using Ward's clustering criterion (Murtagh & Legendre, 2014; Ward, 1963) to visualize the results.

Results and Discussion

The results (shown in Figure 1) revealed that the vast majority of the adjectives (89%) were clustered into their a priori theorized dimensional structure based on findings from the previous three studies. *Negative valence* (the first cluster from the top) was replicated with eight of its adjectives; *humor* (the second cluster) included all of its adjectives as well as three from *positive valence* (sentimental, loving, and lovely); *complexity* (the third cluster) likewise fully replicated and also contained one adjective from *positive valence* (special); *importance* (the fourth cluster) replicated completely; *adversity* (the fifth cluster) replicated with eight of its adjectives; *positive valence* (the sixth cluster) also replicated with six of its adjectives; lastly, *typicality* fully replicated and

added four adjectives to it (tiresome, rowdy, lethal, and overwhelming). The slight differences among the clusters are likely due to the limitations of the distributional hypothesis wherein co-occurrence provides some degree of false-positive associations. Despite this, with a few minor differences, the seven dimensions identified earlier in this article—that is, *negative valence*, *positive valence*, *typicality*, *adversity*, *complexity*, *importance*, and *humor* were replicated using neural-network-based models of natural-language usage.

Phase Three: Conceptual Integration of the Emergent Seven Dimensions into the CAPTION Model

Findings from Phase Two suggested that it may be possible to develop an integrative taxonomic framework for psychological situation characteristics using seven key dimensions which emerged across a diverse array of methodological and analytic lenses. We label this seven-dimensional structure the “CAPTION” model: Complexity, Adversity, Positive Valence, Typicality, Importance, Humor, and Negative Valence. In this section, we provide a brief overview of each dimension, its conceptual content, and its relations with personality manifestation. Note that, while we focus predominately on the implications of these dimensions for personality for the sake of parsimony, these dimensions are broadly derived and, thus, are equally as applicable to a wide-range of content areas (emotions, motivation, character, morality, interests, etc.). We then discuss how these dimensions have emerged in previous situational taxonomic efforts, providing additional support that the lexical approach can integrate both the general aspects of situations and the more conceptually specific components (e.g., frequently occurring situations, personality-relevant situations).

Complexity

The dimension of Complexity asks the question “how complex/intricate is this situation?” There seem to be two major components indicated by this dimension: broadly complex situations and cognitively complex situations. These two components are quite interesting as they are able to capture both generally complex situations, which might include situations that are emotionally or ethically complex, as well as cognitively complex situations, which represents situations marked more by learning, in-depth thought, and investigative exploration. From a personality standpoint, Complexity seems to capture situations in which the personality aspects of openness/intellect, characterized by behaviors such as idea generation/learning (intellect) and introspection/emotional openness (openness), in particular, might manifest (DeYoung, Quilty, & Peterson, 2007; John & Srivastava, 1999; Woo et al., 2014).

Adversity

The dimension of Adversity captures situations that are generally depleting and/or difficult. Situations marked by the exertion of energy, which may or may not be freely chosen, are captured by this dimension. Considering the fact that situations causing stress and resource depletion are such important components of psychological research (Baumeister, Bratslavsky, Muraven, & Tice, 1998; Beehr & Newman, 1978; Lazarus & Folkman, 1984; Muraven & Baumeister,

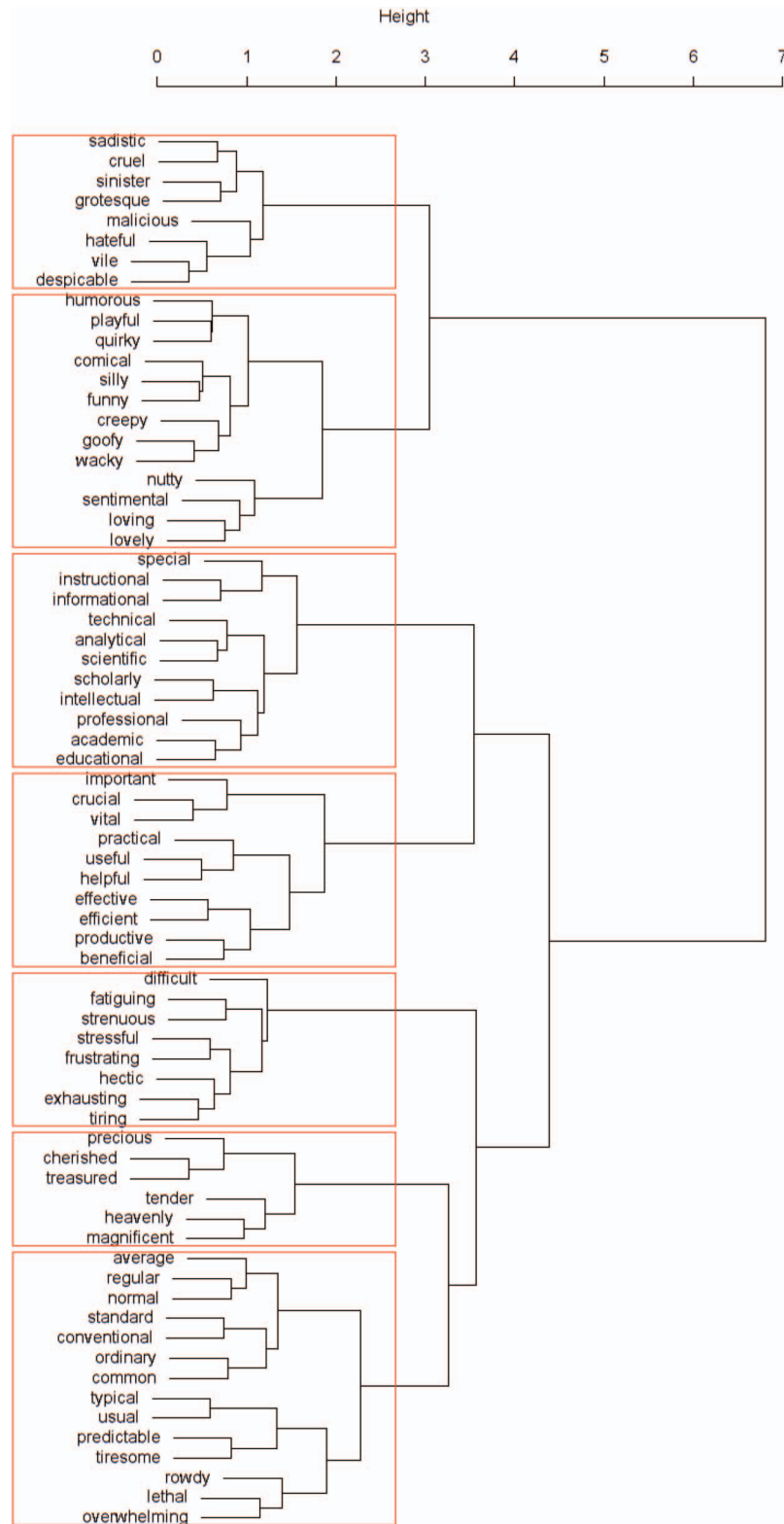


Figure 1. Hierarchical cluster analysis results of the unsupervised vector space representation. Boxes are provided for the seven factor structure for easier visualization of the results. Clusters from top to bottom may be labeled as *negative valence*, *humor*, *complexity*, *importance*, *adversity*, *positive valence*, and *typicality*. See the online article for the color version of this figure.

2000), it is not surprising that this emerges as an important dimension of situations. From a personality perspective, this dimension might be related to contexts in which the personality trait of neuroticism, characterized by frustration, irritation, and anxiety (DeYoung et al., 2007; John & Srivastava, 1999), may manifest. In general, this Adversity dimension seems to capture situations that could be straining for the individual experiencing them.

Positive Valence

The dimension of Positive Valence assesses the degree to which a situation is positively charged. This Positive Valence dimension seems to be composed of two major components: intimacy/interpersonal warmth and general positivity. Whereas the general positivity component captures the positive valence of the situation more generally, the intimacy/interpersonal warmth component seems to represent the positive aspects of the situation as they are related to social love and affection. From a personality standpoint, this dimension could be seen as allowing for the manifestation of both extraversion and agreeableness. The general positivity component can be seen as a situation that might manifest the positive affect-laden and social engagement components of extraversion and the intimacy components might manifest the compassionate aspects of agreeableness (DeYoung et al., 2007; John & Srivastava, 1999). In general, Positive Valence seems to capture the positive valence of the situation through both general positivity as well as intimacy/interpersonal warmth.

Typicality

Typicality is a dimension that measures the commonness and straightforward nature of the situation versus its novelty and ambiguity. From a personality perspective, atypical situations are theorized to be important for the enactment of open and extraverted behaviors, because of the desires of individuals with high levels of these characteristics to encounter unique/novel objects, ideas, persons, and events (DeYoung et al., 2007; Woo et al., 2014). Additionally, Typicality has important implications for the personality-manifestations of conscientiousness. For example, the Orderliness aspect of conscientiousness (DeYoung et al., 2007) might be more likely to be manifested in situations of routine, and the Industriousness aspect might be more likely to be manifested in more straightforward situations, when goal-achievement might be easier (Edwards & Templeton, 2005). Additionally, Typicality also has important implications for other aspects of personality, including personality development, state deviations from trait-typical enactments, and the extent to which behavior might be more automated/scripted (e.g., Buss, 1989; Caspi & Moffitt, 1993; Neyer, Mund, & Zimmermann, 2015; Vazire & Bleidorn, 2015).

Importance

The dimension of Importance generally captures how well the situation might lead to the attainment one's goals. This dimension is in line with the previous literature suggesting that goals are a fundamental component of situations (Yang et al., 2006, 2009). Importance seems to represent both the general importance of situations as well as its direct functionality toward an individual's goals. From a personality standpoint, this dimension could be seen

as situations that might manifest conscientiousness, which is marked by goal-oriented actions, such as achievement striving, focus, and dedication (DeYoung et al., 2007; John & Srivastava, 1999). In general, this dimension assesses situations for their goal-relevance through both their importance and in situations' functionality in achieving them.

Humor

Humor captures the degree to which the situation is humorous or lighthearted. Considering the wealth of research relating to the topic of humor in psychology (e.g., Kuipers, 2015; Martin, 2001, 2010), and the fact that entire industries are dedicated to providing individuals with humorous situations (e.g., comedic movies, TV shows, comedy clubs), it is perhaps unsurprising that this dimension has consistently been found in the literature. An interesting point of note for Humor, is that its valence is mixed, such that it captures both the positive aspects of humorous/lighthearted situations (e.g., situations which are "funny" and "playful"), as well as the negative aspects (e.g., situations which are "juvenile" and "childish"). This likely reflects the fact that humor can be used both constructively (e.g., to alleviate a slightly stressful situation) and destructively (e.g., disparaging/bullying; Kuipers, 2015; Martin, 2010). Thus, although the label "Humor" often connotes positive situations, at its broadest conceptual level, it has a mixed valence.

Negative Valence

Negative Valence assesses situations in regards to how negatively charged they are, such as situations that are potentially malicious or sinister. It is interesting to note that Negative Valence, despite its surface-level opposing relations, does not show evidence of bipolarity with the Positive Valence dimension. Similar to positive and negative affect (Watson, Clark, & Tellegen, 1988), situations often have aspects that are both positive and negative simultaneously. For example, the situation of "being at work" is often somewhat positive (e.g., your interactions with coworkers are wonderful), but also has negative aspects (e.g., customers/clients can be quite rude). Thus, these dimensions are independent dimensions which capture the valence (either positive or negative) present within the situation. It is also useful to note that this dimension was found to be the broadest of all the dimensions included in the CAPTION model, suggesting a rich substructure.

The Ubiquity of the CAPTION Dimensions

The strength of the lexical approach is its ability to integrate the findings from diverse methodological approaches, as it does not limit the content domain of psychological situations to a narrowly specified a priori conceptual focus (e.g., John & Srivastava, 1999; Ten Berge & De Raad, 1999; Yang et al., 2006). Thus, a lexical approach should be able to integrate the dimensions identified across the diverse set of conceptual lenses used in the extant literature. In Table 1 of the supplemental materials we summarize how the previously uncovered dimensions relate to the CAPTION dimensions. As can be seen, the CAPTION dimensions have been found continually in a wealth of studies using differing conceptual lenses, further corroborating the advantages of the lexical approach for integrating diverse lines of taxonomic research. For example,

the broad array of dimensions found across the various approaches (e.g., routine, familiar, organized) are parsimoniously integrated within the broad Typicality dimension within the CAPTION model. In fact, the major dimensions of psychological situations found in the extant literature (see Table 1) can almost entirely be subsumed as components of the CAPTION model, with the exception of the *noisy* dimension and a few, less-frequently replicated components (submissive, with others, etc.). The failure to replicate these dimensions is because these dimensions represent components of the objective situation (i.e., sound, color, lighting, who is present, etc.) or the specific behaviors of actors in the situation (e.g., assertive, submissive), rather than the psychological dimensions of the situation. Therefore, we believe that the CAPTION model represents a comprehensive integration of the psychological situation characteristics domain as previously identified in the literature, and does so at the broadest level of abstraction.

Phase Four: Scale Development and Validation of the CAPTION Dimensions

Based on the lexically derived CAPTION framework, we developed and validated a set of adjective-based scales (the CAPTION Scales or CAPTIONs) to assess these seven dimensions. After developing the initial items, we examined the internal factor structure of the CAPTION model over the course of two studies before conducting investigations around its ability to predict theoretically important outcomes, both in isolation and as compared with other existing situation measures and personality traits. We also create and validate a short-form measure of the CAPTION dimensions.

Item Development

To create the items for assessing each of the CAPTION dimensions, the first three authors of this article discussed the conceptual construct space represented by each of the seven CAPTION dimensions at length and identified 10 items we believed to best represent each dimension. Although the items chosen were generally the strongest-loading items for a given dimension as identified in Phase Two, we also sought to ensure the content breadth covered by the items for each dimension. Through deliberation and iterative group discussions, we developed an initial list of 70 items (10 items for each scale) for capturing the CAPTION dimensions. In the following studies (Study 5 and Study 6), we evaluated whether these items can accurately capture the content of the CAPTION dimensions as described in Phase Three; explored their convergent and divergent validities in relation to extant situation taxonomic frameworks and personality; and examined their abilities to predict important psychological outcomes (e.g., behaviors, affect, motivation, and need satisfaction). The final items resulting from this process can be found in Appendix C.

Study 5: Initial Scale Validation of the CAPTIONs

Method

Data collection. Using a professional service specializing in the customized sampling of survey participants, Qualtrics LLC, we recruited a diverse sample of 506 native English speakers residing

in the United States for participation in our online survey. Participants were, on average, 40.38 years of age ($SD = 12.57$), female (54.35%), and Caucasian (78.7%). Similar to what was done in Studies 2 and 3, the participants were randomly assigned to a time, on the hour, between 8 a.m. and 8 p.m. the day before and were asked to rate the situation they were in at that time. If participants were sleeping, they were asked to rate what they did upon waking. Participants provided a description of the situation they were in at that time and rated each of the measures provided as they applied to that situation. In addition, items measuring participants' self-reported personality were also included (see below).

Measures. The CAPTION dimensions were assessed using the following items: *Complexity*: analytical, academic, scholarly, instructional, complex, technical, intricate, intellectual, scientific, educational; *Adversity*: stressful, fatiguing, frustrating, tiresome, exhausting, tiring, difficult, strenuous, grueling, hectic; *Positive Valence*: heartwarming, cherished, precious, sentimental, loving, affectionate, joyful, special, heavenly, magnificent; *Typicality*: typical, regular, standard, usual, predictable, common, average, normal, ordinary, uneventful; *Importance*: effective, useful, productive, helpful, crucial, important, valuable, vital, beneficial, functional; *Humor*: funny, comical, humorous, silly, goofy, amusing, playful, hilarious, nutty, and wacky; and *Negative Valence*: repulsive, despicable, malicious, grotesque, vile, inhumane, sinister, creepy, sleazy, cruel. In addition to the preliminary CAPTION dimensions, we also included two measures of situational characteristics as points of comparison: (a) one based on van Heck's (1984, 1989) taxonomy of environments and (b) the Riverside Situational Q-Sort 8 (RSQ-8; Rauthmann et al., 2014). Van Heck's (1984, 1989) model of situation classes is predominately a taxonomy of the environments that situations take place within. The situation classes identified are those of interpersonal conflict, joint working and information exchange, intimacy and interpersonal relations, recreation, traveling, rituals, sport, excesses, serving, and trading. Each dimension was measured using an item created based upon van Heck's discussion of these dimensions, for a total of 10 items. The RSQ-8 (Rauthmann et al., 2014) measures eight dimensions of psychologically important personality-relevant contexts: Duty, Intellect, Adversity, Mating, positivity, Negativity, Deception, and Sociality - also known as "DIAMONDS." The DIAMONDS dimensions were derived by factor analyzing 89 contexts which might manifest personality-relevant behaviors derived from the California Adult Q-Sort personality scale (Block, 1961). Reliabilities for the 32-item RSQ-8 were acceptable ($\alpha s = .79-.93$). We chose to include these two particular measures because they are recognized as two of the most useful situation taxonomies to date (Rauthmann et al., 2014; Ten Berge & De Raad, 1999) and also represent two of the other major approaches to situational taxonomic development (van Heck's model captures a set of environments and the DIAMONDS dimensions describe personality-relevant contexts). To assess personality, we used Goldberg's (1990) 50-item Big Five Factor Marker scale from the International Personality Item Pool (Goldberg, 1999). The inventory showed acceptable reliabilities ($\alpha s = .87-.93$).

Results and Discussion

Internal factor structure. To establish the psychometric soundness of the initial CAPTIONs items, we used exploratory

structural equation modeling (ESEM; Asparouhov & Muthén, 2009) in Mplus (Muthén & Muthén, 1998-2012). ESEM was chosen for its ability to overcome the untenable assumptions of confirmatory factor analysis for complex psychological constructs (i.e., multidimensional scales with more than 50 items; Asparouhov & Muthén, 2009; Marsh et al., 2009, 2010), which tends to cause distorted factors and factor correlations (Asparouhov & Muthén, 2009). Indicators of many psychological constructs lie between the interstitial space of factors, and the requirement of zero cross-loadings is too strict (Hopwood & Donnellan, 2010). This issue becomes much more pronounced in complex psychological measures. ESEM relaxes this requirement by allowing low cross-loadings, which has a greater fidelity to complex psychological measures. Furthermore, unlike principal components analysis, ESEM provides the full functionality of structural equation modeling (e.g., model fit indices, correlated uniquenesses).

The ESEM results (see Table 6) revealed a clear seven-dimensional structure. The primary loadings were unequivocally strong, with negligible cross-loadings (none above .30 and only two between .20 and .29). The indices of fit further supported these findings, with the seven-dimensional structure fitting the data well ($\chi^2 = 6189.23$, $df = 1946$, CFI = .90, TLI = .87, RMSEA = .07, and SRMR = .02). The scale reliabilities for the CAPTION dimensions, which can be found in the top of Table 7, were all also very strong, with Cronbach's alpha ranging from .94 to .99. Together, the clear conceptual interpretability of the seven-dimensional CAPTION model, along with the empirical fit indices, supports the validity of the CAPTIONs in measuring the theorized seven-dimension structure. The only point of concern with the current measurement structure was the substantial positive correlation between Humor and Positive Valence ($r = .58$), with notably smaller relations found between Humor and Negative Valence ($r = .32$). This pattern is dissimilar to the factor correlations in the dimension identification stage where Humor was more strongly correlated with Negative Valence than with Positive Valence. Thus, it seems that the initial items chosen to assess Humor more strongly reflect the positive aspects of the Humor dimension (playful, amusing, etc.) rather than its more negative aspects (childish, juvenile, etc.). This deficiency is rectified in Study 6 (and in the final CAPTIONs items located in Appendix C), where both aspects of the Humor factor are represented in the items used.

Relations with other situation measures. Descriptive statistics and bivariate correlations of the CAPTION dimensions with the other two situation measures (i.e., van Heck's and the RSQ-8) can be found in Table 7. The results indicate coherent patterns of relations between the CAPTIONs and van Heck's dimensions. For example, situation classes involving interpersonal conflict were the most strongly related with Negative Valence, suggesting that relational conflict can be potentially harmful and concerning to individuals. This makes sense as situations of interpersonal conflict might interfere with the satisfaction of the need for relatedness/belonging (Baumeister & Leary, 1995; Ryan & Deci, 2000)—increasing the likelihood for individuals to perceive those situations as more negative. Similarly, Recreation was found to be positively associated with both Positive Valence and Humor. Considering the fact that leisure engagement fulfills a wide-variety of needs such as affiliation and autonomy (e.g., Kuykendall, Tay, & Ng, 2015; Newman, Tay, & Diener, 2014), it makes sense that leisure activities would be experienced as more positive and light-

hearted. Overall, the pattern of relations reveals a coherent manner supporting the convergence of these two dimensional conceptualizations.

Similar patterns of relations with the RSQ-8 were found and, as would be expected because of the stronger degree of conceptual overlap between the CAPTION and the DIAMONDS models, the CAPTION dimensions (measured with the CAPTIONs) generally showed stronger correlations with the DIAMONDS dimensions (measured with the RSQ-8) than with van Heck's. For example, the RSQ-8's Duty factor mapped relatively well onto the Importance factor of the CAPTIONs. This pattern makes sense as the Importance dimension of the CAPTION model captures both the importance and the goal-relevance of the situation. As important and goal-relevant situations often necessitate the perceived behavioral requirements of "task-oriented thinking is needed" or that "focusing on details is required," it makes sense that these two dimensions would show positive interrelations.

Although many of the RSQ-8 factors show moderate to strong relations with a single CAPTIONs factor in a theoretically interpretable manner (e.g., Intellect with Complexity, Positive Valence with pOsitivity), three RSQ-8 factors (Adversity, Mating, and Deception) showed substantial overlap with the Negative Valence factor of the CAPTIONs (r s ranging from .52 to .64), as well as among one another (r s of .68, .60, & .44). Based on a closer examination of the scales' item contents, and of the Negative Valence-loading adjectives in Studies 2 and 3, we viewed this pattern of correlations as reasonable, considering that the specific situations of being antagonized/blamed (Adversity), sexual/physical overtones (Mating), and hostility/deceit (Deception), as construed as DIAMONDS dimensions, can be conceptualized as falling under the breadth of the Negative Valence dimension within the CAPTION framework. In line with this, our follow-up analysis revealed that adjectives from Study 2 relating to these three DIAMONDS dimensions (Adversity, Mating, and Deception) all loaded strongly, and cleanly, onto the broad CAPTION Negative Valence dimension, as can be seen in Table 8. Thus, both theoretically and empirically, one could understand these three DIAMONDS dimensions as possible facets of Negative Valence, as broadly conceptualized within the CAPTION framework. Although one could argue that, as "bad" is often stronger than "good" (e.g., Baumeister, Bratslavsky Finkenauer, & Vohs, 2001), having additional dimensions for Negative Valence could be understandable, the empirical findings from the 851 and 535 adjective lists suggest that this dimension is best represented at a higher level of abstraction. Explorations of the lower-level components of Negative Valence, as well as all of the CAPTION dimensions, will clearly be important for future research, just as they have been for lexically derived personality structures (e.g., Costa & McCrae, 1995; DeYoung et al., 2007; Goldberg, 1999). Taken together, these findings support the conclusion that the CAPTION framework likely represents the broadest conceptualization of situational characteristics by including both the general and the more specific characteristics (e.g., personality-relevance) of situations.

The pattern of relations found between measures of the DIAMONDS Mating dimension and the CAPTION Negative Valence dimension warrants further clarification. Our data indicate that people typically view situations of Mating ("Situation includes stimuli that could be construed sexually," "racy," etc.) in a negative manner, possibly because situations involving unwanted sex-

Table 6
Exploratory Structural Equation Modeling of the Initial CAPTIONs Items

Adjective	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
academic	.89	.02	.08	.06	-.13	.00	-.01
scholarly	.86	-.04	.13	.02	-.10	-.04	.04
intellectual	.84	.00	-.02	.00	.02	.02	-.04
educational	.76	-.03	.08	-.01	-.02	.06	-.03
scientific	.74	.05	.10	.02	-.02	.02	.03
technical	.70	.10	-.21	.02	.08	.00	.05
analytical	.69	.04	-.18	.01	.12	.03	.02
informational	.67	-.07	-.04	-.01	.11	.08	-.03
intricate	.65	.23	.00	-.05	.10	-.06	-.02
complex	.63	.27	-.09	-.09	.02	.00	.02
exhausting	-.08	.97	.05	-.02	.00	.01	-.02
tiring	-.09	.94	-.03	.00	.06	.02	-.04
fatiguing	-.06	.93	.03	.01	-.01	.00	.01
tiresome	.01	.89	-.01	.07	-.02	-.01	.02
stressful	.05	.82	-.07	-.03	.00	-.03	.03
strenuous	.05	.79	.05	.01	.02	.02	-.01
hectic	.10	.77	.03	-.05	.04	.06	-.04
frustrating	.08	.75	-.12	.02	-.04	.03	.05
difficult	.06	.74	.01	-.09	.05	-.05	.01
grueling	.15	.72	.01	.04	-.05	-.04	.11
precious	.03	.02	.92	.04	-.01	-.02	.01
loving	-.05	.06	.89	.03	.02	.02	-.01
heartwarming	-.06	.06	.89	-.01	-.01	.05	-.07
special	.02	.03	.89	-.02	.03	-.02	.00
affectionate	-.01	.05	.87	.01	-.03	.02	.01
cherished	-.04	.01	.86	.00	.03	.04	-.01
heavenly	.04	-.07	.81	.02	.04	-.05	.10
sentimental	.06	.00	.76	-.03	-.02	.10	.07
magnificent	.10	-.10	.75	-.07	.04	.02	.09
joyful	.00	-.13	.74	.01	.00	.10	-.02
regular	-.01	-.01	.02	.87	.05	.01	-.03
normal	.04	-.05	.05	.83	-.01	-.05	-.03
standard	.02	.02	-.01	.83	.06	-.03	.01
ordinary	.01	-.01	-.06	.82	-.06	.00	-.01
average	.03	-.01	.00	.81	-.02	.06	.02
common	.00	.00	.03	.78	.02	-.02	-.01
usual	.05	-.01	.00	.78	.03	-.02	-.06
typical	-.03	.00	-.01	.75	.00	.03	-.01
predictable	-.07	.08	-.02	.71	.15	.05	.06
uneventful	-.04	-.02	-.05	.60	-.05	-.08	.08
useful	.00	-.06	-.03	-.01	.88	-.06	.01
effective	-.04	-.03	-.01	.04	.85	.06	.02
beneficial	-.04	-.05	.04	-.01	.83	-.01	-.05
productive	.04	.05	-.04	-.05	.80	-.05	-.05
vital	.00	.07	.02	.02	.77	.02	.03
important	.01	.08	.06	-.01	.76	-.07	.03
valuable	.02	-.02	.16	-.02	.74	.03	-.04
helpful	.12	-.02	.04	.08	.69	.02	.01
functional	.05	.02	.01	.10	.68	.01	-.01
crucial	.02	.17	-.04	.05	.60	.00	.05
silly	-.04	.08	-.04	.01	-.02	.96	-.03
goofy	-.04	.04	.00	.00	.00	.92	.04
humorous	.05	-.05	.02	.01	.00	.91	-.01
comical	.03	-.02	.03	.02	-.01	.91	.00
funny	.03	-.04	.03	-.02	.00	.90	-.01
hilarious	.01	.01	.04	-.01	.02	.88	.01
wacky	-.02	.06	-.03	.02	-.04	.87	.09
nutty	-.02	.07	-.02	.02	-.02	.86	.10
amusing	.08	-.10	.07	-.03	.03	.81	-.02
playful	.01	-.05	.19	-.04	.03	.77	-.01
repulsive	-.03	.03	.00	.01	.01	.00	.94
despicable	.00	.00	.00	-.01	.01	-.01	.94
sinister	.02	-.01	.02	-.01	-.01	-.01	.94

(table continues)

Table 6 (continued)

Adjective	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
malicious	-.01	.02	.04	-.02	.00	-.02	.94
grotesque	.00	-.02	.02	.01	.02	.02	.93
creepy	.02	-.04	.02	.01	.00	.05	.92
sleazy	.02	-.02	.01	.03	-.01	.04	.92
cruel	.00	.01	-.02	-.04	-.03	-.01	.92
inhumane	.03	.02	.01	.01	-.04	-.01	.90
vile	-.02	.05	-.06	.00	.04	.07	.86

Note. $n = 506$. The strongest loadings for each factor are bolded.

ual advances and/or purely surface-level physical/sexual encounters are likely construed as negative (e.g., “sleazy,” “creepy,” “repulsive”). However, this should not be taken to suggest that *all* sexual encounters are experienced as negative. In fact, adjectives that are often used to characterize sexual situations involving deeper-level personal connections (e.g., “intimate,” “flattering,” “passionate,” “affectionate”) all showed strong and clean loadings

onto the CAPTION Positive Valence dimension in our data ($\lambda_s = .54-.80$). Taken together, these findings suggest that while predominantly “sexual” situations are usually experienced as negative, those signaling psychological (as well as physical) intimacy are often perceived as positive.

We make one additional (yet important) note regarding the relation between the CAPTION and the DIAMONDS models: The labeling of

Table 7

Descriptives and Intercorrelations for Study Variables in Phase 4: Study 5

Variable	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
<i>M</i>	2.62	2.54	2.64	3.72	2.57	2.40	1.42
<i>SD</i>	1.04	1.14	1.11	.90	.92	1.11	.76
CAPTIONs							
Complexity	(.94)						
Adversity	.42*	(.96)					
Positive Valence	.18*	-.11*	(.97)				
Typicality	.07	-.03	-.07	(.94)			
Importance	.37*	.27*	.25*	.24*	(.94)		
Humor	.22*	-.02	.58*	-.02	.01	(.98)	
Negative Valence	.29*	.31*	.20*	.06	.01	.32*	(.99)
van Heck's dimensions							
Interpersonal conflict	.33*	.30*	.19*	.03	.10*	.28*	.53*
Joint working ^a	.42*	.29*	.18*	-.02	.27*	.27*	.22*
Intimacy ^b	.26*	.13*	.32*	.00	.07	.37*	.58*
Recreation	.07	-.07	.37*	.00	-.03	.40*	.18*
Traveling	.20*	.26*	.22*	-.01	.13*	.20*	.37*
Rituals	.27*	.17*	.36*	.02	.08	.29*	.50*
Sport	.28*	.20*	.35*	-.01	.17*	.35*	.38*
Excess	.27*	.15*	.33*	-.04	.02	.36*	.56*
Serving	.25*	.21*	.29*	.11*	.30*	.18*	.26*
Trading	.33*	.23*	.25*	.03	.14*	.34*	.45*
RSQ-8 (DIAMONDS)							
Duty	.42*	.47*	-.08	.12*	.61*	-.12*	.06
Intellect	.64*	.35*	.14*	.03	.38*	.15*	.25*
Adversity	.39*	.39*	.17*	.03	.10*	.28*	.64*
Mating	.24*	.12*	.35*	.00	.12*	.38*	.52*
pOsitivity	.01	-.21*	.54*	.03	.15*	.54*	.08
Negativity	.37*	.63*	-.09*	.03	.21*	.04	.46*
Deception	.34*	.40*	.02	.04	.09*	.19*	.58*
Sociality	.21*	.16*	.20*	.01	.14*	.31*	.23*
Personality							
Extraversion	.13*	.01	.12*	-.11*	.13*	.09*	-.01
Agreeableness	-.06	-.10*	.00	.03	.20*	-.07*	-.32*
Conscientiousness	.05	-.02	-.05	.05	.27*	-.15*	-.17*
Neuroticism	-.07	.18*	-.07	.00	-.11*	-.02	.18*
Openness	.11*	-.03	.00	.09*	.21*	.02	-.17*

Note. $n = 506$.

^a Joint working and information exchange. ^b Intimacy and interpersonal relations.

* $p < .05$.

Table 8

Mapping the DIAMONDS Negative Valence-Related Dimensions to Adjectives From Study 2

Adjective	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
Adversity							
lethal	-.03	-.19	.01	.00	.15	-.11	.92
menacing	-.01	-.06	-.06	.01	.15	-.01	.90
hostile	.01	-.06	-.09	-.01	.07	-.03	.88
threatening	-.02	-.05	-.05	-.02	.13	-.02	.88
Mating							
steamy	.02	-.26	.10	.08	.10	-.02	.70
racy	.05	-.25	.00	.02	.01	.17	.67
sexual	.07	-.32	.12	-.04	.02	.02	.59
flirtatious	.03	-.24	.16	-.01	-.01	.14	.52
Deception							
dishonest	-.05	-.12	-.02	.03	.01	-.04	.86
deceptive	.09	-.11	-.08	.05	.00	.02	.74
unsportsmanlike	-.02	-.04	-.04	.10	.04	.07	.72
phony	.02	-.04	-.08	.06	-.13	.02	.70

Note. $n = 1,689$. The strongest loadings for each factor are bolded.

the Negative Valence and Adversity dimensions in the CAPTION model do not align with those in the DIAMONDS model (see Table 8). This was done because our data showed that, within the CAPTION framework, Negative Valence represents a broad dimension of negatively charged situations, whereas Adversity represents a much narrower dimension primarily associated with depleting/effortful situations. In Studies 2 and 3, Negative Valence emerged as the broadest factor (e.g., 290 adjectives in Study 2 and 166 adjectives in Study 3 with loadings above .50) and was characterized by items representing negative valence (e.g., “negative,” “harmful,” “bad,” “vile,” “cruel”) in the final seven- and six-factor solutions. This pattern remains relatively unchanged from when only two broad, higher-order (i.e., positive and negative valence) factors were extracted from the data (similar to two-factor solutions for personality; e.g., Saucier & Srivastava, 2015). The Adversity factor, on the other hand, subsumed a much smaller number of adjectives (e.g., 17 adjectives for Study 2 and 12 adjectives for Study 3) and was characterized by items that exclusively related to how depleting/effortful the situation is (e.g., “exhausting,” “tiresome,” “strenuous,” “exasperating”) in our factor analytic results. While the DIAMONDS dimension representing encountering blame/criticism is currently labeled as Adversity within their framework, it might actually be better labeled as Antagonism. Regardless, the Negative Valence and Adversity dimensions within the CAPTION taxonomy represent dimensions of broad negative valence and of depleting/effortful situations, respectively.

Relations with personality. Overall, the CAPTION dimensions were related to personality traits in a theoretically coherent manner. Experiencing a “complex” situation was found to be related with openness to experience and extraversion. This finding is in line with the definition of openness, describing the tendency of individuals who seek out novel and complex stimuli (DeYoung et al., 2007; Woo et al., 2014). Similarly, extraverts tend to engage in risky behaviors (e.g., Hoyle, Fejfar, & Miller, 2000; Schmitt, 2004), which could increase the likelihood of encountering ethically or emotionally complex situations. Experiencing an “adverse” situation was found to be positively related with neuroticism and negatively related with agreeableness. As neurotic

individuals are more likely to be anxious and stressed, it is likely that they might be more likely to perceive and/or experience situations to be more challenging. In contrast, more agreeable individuals may be more likely to see situations, and the individuals within them, to be more kind and relaxing. The experience of a “positive” situation was related with extraversion, as would be expected due to its consistently found relations with positive affect (e.g., Costa & McCrae, 1980). The exploration-oriented personality characteristics of extraversion and openness/intellect were also found to be related with the experience of situations characterized by Typicality. Although the opposing pattern of relations of these two traits is somewhat surprising, it might be that highly open individuals, characterized by their desire for novelty, may set a higher bar for what constitutes as a novel (i.e., atypical) situation, whereas extraverts, due to their tendency toward higher positive affect, may tend to perceive more novel opportunities within the situations they experience. Importance was associated with all of the Big Five traits (positively, with the exception of neuroticism), with conscientiousness showing the strongest correlation ($r = .27$). Considering the industriousness of conscientious individuals, it makes sense that they would experience situations that are more important/productive in nature. The experience of situations as “humorous” was also most strongly related to conscientiousness ($r = -.15$). This is an interesting finding, as it suggests that highly conscientious individuals may be more practical and task-focused, and thus less likely to encounter and/or perceive situations as humorous. Lastly, the experience of a “negative” situation was most strongly related to agreeableness ($r = -.32$). As highly agreeable individuals are often more compassionate and polite (DeYoung et al., 2007), they might be less likely to encounter and/or perceive situations of negativity.

Summary. Taken together, these findings provide initial validity evidence for the CAPTIONs. We found evidence for the seven-dimensional structure as assessed by the 70-item scale. The results from Study 5 also support the notion that a lexically derived taxonomy captures the broadest-possible set of descriptors available and, therefore, can integrate the findings from taxonomies derived using differing conceptual lenses (e.g., John & Srivastava, 1999; Ten Berge & De Raad, 1999; Yang et al., 2006). One

particular concern regarding the findings of Study 5 pertained to the coverage of the negative aspects of the *Humor* factor; we sought to address this issue in Study 6 by revising the measure.

Study 6: Further Validation of the CAPTIONs

Method

Findings from Study 5 provided strong evidence that the initial CAPTIONs items were generally able to assess the broad content domain of psychological situation characteristics. However, we also identified some important limitations in the assessment of the *Humor* dimension: inadequate coverage of the negative aspects of *Humor*. To rectify this concern, we conducted an additional study in which we more accurately assessed the full breadth of this dimension. In addition to revising/finalizing the CAPTIONs items, we also sought to provide some initial predictive validity evidence by assessing its relations with a broad array of outcomes (in situ behaviors, affect, motivation, and need satisfaction). We assess the predictive validity of the CAPTIONs in three ways: considering the seven CAPTION dimensions as (a) independent predictors (i.e., each dimension, in isolation), (b) as a set of predictors (i.e., the CAPTION dimensions, as a group), and (c) in relation to other situation measures. Lastly, we created and validated a shortened version of CAPTIONs—consisting of 28 items, and assessed its convergence with the full-length version as well as its predictive validity.

For a situational taxonomy to be useful, it must be able to predict important psychological outcomes, such as the behaviors, affect, and other psychological outcomes occurring within a situation (Funder, 2006; Rauthmann et al., 2014; Reis, 2008; Ten Berge & De Raad, 2002; Yang et al., 2009). To this end, we sought to explore the CAPTION model's predictive validity in regards to in situ accounts of personality-relevant behaviors, affect, need satisfaction, and intrinsic motivation. These outcomes were chosen as they represent a broad-array of some of the most fundamental outcomes of interest in psychology. As the CAPTION model was derived with the intention of capturing the full lexical domain (e.g., John & Srivastava, 1999; Ten Berge & De Raad, 1999; Yang et al., 2006), it should be able to capture the dimensions of situations which are generally psychologically important, as well as situations which are more specific. Because of the breadth of the CAPTION dimensions, we expect the CAPTIONs, as a whole, to predict a broad range of psychological outcomes. At the same time, we also expect that the CAPTIONs in its current form (without specifying the more fine-grained, lower-level facets within each dimension) may not capture some meaningful variance which may be explained by other existing measures designed to capture more specific situation characteristics (e.g., the multiple Negative Valence-related components of the DIAMONDS taxonomy).

Hypothesized relations with outcomes. For personality-relevant behaviors in situ, we draw on the conceptual content of the CAPTION situation dimensions (see Phase Three), as well as the construct space of personality (e.g., Goldberg, 1990; John & Srivastava, 1999) to inform our predictions. We predicted that (situational) Complexity would be positively associated with exhibiting open/intellectual and conscientious behaviors, as these situations would likely entail complex intellectual and/or emo-

tional discourse which would be more likely to require the engagement of analytic and detail-oriented thinking; Adversity would be positively related to neurotic behaviors, as situations involving stress and frustration might lead to one being nervous, angry, and tense; Positive Valence, because of its capturing of intimate and positive situations, would be positively related to agreeable and extraverted behaviors and negatively with neurotic behaviors, as these situations would likely involve connecting with others (agreeable and extraverted behaviors) and calming behaviors (i.e., neurotic, reverse); Importance would be positively related to conscientious behaviors, as important situations would increase the likelihood of the desire for achievement-oriented action such as organized, thorough, and hardworking behaviors; *Humor* would be positively related to agreeable and extraverted behaviors, as humorous situations frequently push for in-group-oriented behaviors (Kuipers, 2015; Martin, 2010), leading to increased energy, cooperation, and boldness; and Negative Valence would be positively related to neurotic and negatively to agreeable behaviors, as situations characterized by deception, creepiness, and cruelty would call for more withdrawal actions, such as showing more discontent and/or being less cooperative.

For affect, we predicted that positive affect would be most likely experienced in situations with high levels of Positive Valence and *Humor*, and least likely in situations with Adversity and Negative Valence. As positive and humorous situations can frequently serve to fulfill individuals' need for belonging/relatedness (Kuipers, 2015; Martin, 2010), we expect that positive affect will be increased in response to this need satisfaction. As adverse and negative situations may thwart the potential satisfaction of needs, such as belonging, meaningful existence, self-esteem, and control, we posit that these situations will likely be less likely to elicit positive affect. The opposing pattern of relations was anticipated for negative affect. For need satisfaction, we expected that the need for belonging would be most satisfied in situations with high Positive Valence and least satisfied in situations with high Negative Valence, as these two situations entail potential interpersonal intimacy and harm, respectively; the need for self-esteem would be most satisfied in situations with high levels of Positive Valence and Importance, while negatively affected in situations with Negative Valence and Adversity, as these situations entail affirmation or challenges to the self (i.e., Positive and Negative Valence) and/or the goals one has (i.e., Importance, Negative Valence, and Adversity); the need for meaningful existence would be most satisfied in situations with high levels of Importance, Complexity, and Positive Valence, while negatively affected in situations with Negative Valence and Typicality, as these situations would either support or hinder the ability to derive satisfaction from interpersonal relationships (i.e., Positive and Negative Valence) or the goals one is trying to achieve (i.e., Importance, Typicality, and Complexity); and the need for control would be most satisfied in situations with Importance and Typicality, and least satisfied in situations with Negative Valence, as these situations would more readily allow for one to pursue desired goals by providing opportunities and consistencies for their success (i.e., Importance and Typicality) or hindrances thereof (i.e., Negative Valence and Adversity). For intrinsic motivation, we predicted that the situations with high levels of Positive Valence, Importance, and *Humor* would be positively associated with intrinsic motivation, and that situations with high Negative Valence would be negatively related.

Data collection. Using the same methods as Study 5, we recruited another diverse sample of 522 native English speakers residing in the United States for participating in our online survey. Participants were, on average, 39.97 years of age ($SD = 12.55$), female (50.04%), and Caucasian (74.3%). The survey procedures were the same as in Study 5.

Measures. We assessed *situations* in the same manner as in Study 5 (using the CAPTIONs items, van Heck's dimensions, and the RSQ-8), with two notable exceptions. First, we revised the Humor scale items in order to more accurately reflect this dimension (by ensuring items assessing both the positive and negative aspects of Humor were included). The final items of the Humor scale are funny, silly, goofy, nutty, wacky, mischievous, crazy, juvenile, immature, and childish. Second, based on Edwards and Templeton's (2005) work, we added another set of situational adjectives to be compared with the CAPTIONs, to further exploring the CAPTIONs' convergent validity. While Edwards and Templeton found differences in the dimensions both within and across their studies, making the interpretations of the dimensions quite challenging, they did discuss several items that were related to the dimensions which occurred the most frequently across the word lists used (*positivity*: exciting, awesome, outstanding; *negativity*: obscene, negative, polluted; *productivity/goal achievement*: productive, orderly, vital; and *routineness/ease of negotiation*: simple relaxed, and carefree). Reliabilities for all of the situation conceptualizations were acceptable (CAPTIONs: $\alpha s = .94-.98$; the RSQ-8: $\alpha s = .79-.94$; Edwards and Templeton: $\alpha s = .77-.93$).

We assessed *personality-relevant behaviors* using an adapted version of Goldberg's bipolar adjective scale (Goldberg, 1990), consistent with what has been done in previous research assessing behaviors in various situations (e.g., Fleeson, 2001, 2004). Participants were asked to rate how well each of the bipolar scales best described their in situ behaviors using a 5-point bipolar rating scale. For example, the participant would rate how "talkative" versus "silent" they were in the situation. Reliabilities for personality-relevant behaviors were acceptable ($\alpha s = .85-.89$). *Positive and Negative Affect* were measured using the Positive and Negative Affect Schedule (Watson et al., 1988), asking the participant to rate their affect experienced in situ. Reliabilities were acceptable ($\alpha = .90$ & $.95$, respectively). *Need satisfaction* was assessed using 18 items measuring in situ need satisfaction of the four core needs for belonging, self-esteem, meaningful existence, and control which have been used with success in numerous social psychological studies (e.g., van Beest & Williams, 2006; Williams, Cheung, & Choi, 2000; Zadro, Williams, & Richardson, 2004). Example questions include: "I felt a sense of belonging," "My self-esteem was high," "I felt meaningless" (reverse scored), and "I felt I had control over the situation" for each need, respectively ($\alpha s = .52-.79$). *Intrinsic motivation* was measured using a seven-item intrinsic motivation scale adapted from the Intrinsic Motivation Inventory (Ryan, 1982). Example items include "This situation was quite enjoyable" and "This situation did not hold my attention at all" (reverse scored). The Intrinsic Motivation Inventory was found to have good reliability ($\alpha = .88$). Lastly, we use the HEXACO 60-item personality measure (Ashton & Lee, 2009) to assess *personality* to compare the CAPTIONs' predictive validity relative to this broader, six-factor conceptualization of personality ($\alpha s = .69-.81$).

Results and Discussion

Internal factor structure. To validate the psychometric soundness of the final set of CAPTIONs items, we used exploratory structural equation modeling (ESEM; Asparouhov & Muthén, 2009) in Mplus (Muthén & Muthén, 1998-2012). The ESEM results (in Table 9) revealed a clear seven-dimensional structure. Similar to Study 5, each of the primary loadings were quite strong, with negligible cross-loadings. The indices of fit further supported these findings, with the seven dimensional structure fitting the data relatively well ($\chi^2 = 6680.92$, $df = 1946$, CFI = .89, TLI = .86, RMSEA = .07, and SRMR = .02) compared with what is typically seen in personality taxonomic research (e.g., Marsh et al., 2010). The scale reliabilities for the CAPTION dimensions were all very strong, with Cronbach's alpha ranging from .94 to .98 (see Table 10). Together, the clear interpretability of the seven-dimensional CAPTION model, along with the empirical fit indices, supports the validity of the CAPTIONs in assessing the theorized seven dimensional structure. Additionally, the Humor scale now more accurately represents both the positive and negative aspects of this dimension.

Relations with situation measures. The pattern of relations between the CAPTIONs and the other situation measures were largely the same as in Study 5. Additionally, the Edwards and Templeton (2005) items included in the current study provide further evidence for the validity of the CAPTION dimensions, with the strongest relations among the situation dimensions being with their theoretically matched components (Positive Valence-Positivity $r = .73$; Negative Valence-Negativity $r = .79$; Importance-Productivity/Goal Achievement $r = .70$). Not surprisingly, the dimension which showed the least consistency in the Edwards and Templeton (2005) paper, the Routineness/Ease of Negotiation dimension, did not relate as cleanly with the CAPTION dimensions as the three more consistently emerging Edwards & Templeton dimensions (i.e., Positivity, Negativity, and Productivity/Goal Achievement). As can be seen across the word lists used (Table 1 in Edwards & Templeton, 2005), as well as in the name of the dimension itself, the Routineness/Ease of Negotiation factor appears to be representative of two CAPTION dimensions: Typicality (Routineness) and Adversity (Ease of Negotiation, reversed). Thus, despite the limited content coverage of situation characteristics within each word list analyzed (i.e., random subsets of 100–114 adjectives were used), the Edwards and Templeton study identified all of the CAPTION dimensions except for Humor and Complexity, albeit somewhat inconsistently. This remarkable consistency provides additional support that the CAPTION dimensions capture the full-breadth of the content domain of psychological situation characteristics.

Relations with outcomes. Most of the hypotheses regarding in situ behaviors were supported, with several associations which were not hypothesized also emerging. For example, in addition to positive situations, extraverted behaviors were also more likely to be engaged in during complex situations, which might be attributable to the need for an increased need to communicate to work through such situations. Positive situations were more likely to be related with exploratory behaviors (extraversion and openness) as well as agreeable behaviors, suggesting that in positive situations individuals may feel freer to engage in more exploratory and/or prosocial behaviors. In addition to predicting conscientious behav-

Table 9
Exploratory Structural Equation Modeling of the CAPTIONs Measure

Adjective	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
analytical	.69	.04	-.10	.00	.06	.04	.01
complex	.58	.32	.00	.00	.03	.00	.00
technical	.64	.17	-.10	.02	.04	.06	.00
intricate	.57	.27	.01	.00	.03	.03	.00
academic	.81	.01	.00	.00	.00	.00	.09
intellectual	.87	.00	.04	.05	.00	.00	.00
scholarly	.85	.00	.01	.02	.00	.02	.09
scientific	.71	.01	.04	.00	.00	.00	.20
educational	.82	.00	.04	.00	.01	.00	.00
instructional	.72	.06	.03	.00	.09	.00	.00
stressful	.06	.87	.00	.00	.00	.00	.00
exhausting	.00	.95	.02	.02	.00	.00	.00
tiring	.00	.94	.02	.04	.00	.00	.00
difficult	.11	.77	.04	.00	.00	.00	.07
fatiguing	.00	.87	.00	.01	.01	.02	.04
strenuous	.11	.76	.00	.01	.00	.03	.05
frustrating	.03	.80	.00	.00	.00	.00	.02
grueling	.15	.67	.02	.00	.00	.01	.16
tiresome	.00	.87	.00	.02	.05	.02	.00
hectic	.00	.73	.02	.00	.04	.09	.04
loving	.00	.03	.86	.00	.06	.00	.02
heartwarming	.00	.01	.89	.00	.01	.00	.00
cherished	.00	.00	.91	.01	.02	.02	.00
joyful	.00	.00	.83	.01	.00	.05	.00
precious	.00	.02	.92	.02	.00	.00	.00
special	.01	.00	.88	.00	.03	.00	.01
heavenly	.04	.00	.85	.04	.01	.00	.13
magnificent	.08	.00	.80	.03	.00	.05	.09
sentimental	.00	.07	.81	.00	.00	.07	.02
affectionate	.02	.05	.84	.00	.00	.07	.02
typical	.07	.00	.00	.71	.04	.00	.01
predictable	.00	.06	.00	.69	.05	.00	.05
common	.03	.00	.04	.84	.00	.00	.00
average	.00	.00	.00	.78	.00	.08	.03
regular	.00	.01	.03	.88	.03	.00	.00
standard	.02	.00	.06	.86	.01	.00	.01
normal	.01	.00	.06	.85	.01	.00	.00
usual	.00	.00	.00	.82	.01	.01	.00
ordinary	.00	.03	.00	.85	.00	.04	.00
uneventful	-.10	.02	-.10	.51	.00	.00	.12
crucial	.00	.19	.00	.07	.51	.00	.13
important	.01	.13	.09	.00	.75	.00	.00
effective	.04	.02	.03	.06	.79	.00	.00
valuable	.08	.00	.22	.00	.72	.00	.00
vital	.00	.06	.08	.03	.61	.01	.06
useful	.00	.00	.00	.01	.89	.00	.06
beneficial	.00	-.10	.00	.00	.89	.00	.01
productive	.03	.01	.00	.00	.87	.00	.00
helpful	.04	.00	.02	.00	.79	.07	.00
functional	.06	.02	.00	.09	.73	.04	.00
funny	.00	.00	.28	.00	.00	.67	.00
silly	.00	.02	.13	.00	.00	.87	-.10
goofy	.01	.02	.15	.00	.01	.85	.00
nutty	.00	.00	.02	.01	.00	.90	.04
wacky	.02	.00	.00	.00	.02	.89	.04
mischievous	.00	.00	.07	.00	.01	.71	.12
crazy	.06	.14	.00	.00	.03	.73	.06
juvenile	.03	.05	.00	.00	.00	.65	.20
immature	.02	.04	.00	.03	.00	.68	.21
childish	.00	.06	.02	.01	.00	.71	.08
vile	.02	.11	.03	.00	.00	.00	.80
inhumane	.00	.01	.01	.01	.00	.00	.89
sinister	.00	.00	.01	.00	.01	.00	.96
creepy	.00	.00	.00	.00	.02	.04	.92

(table continues)

Table 9 (continued)

Adjective	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
repulsive	.00	.00	.00	.00	.01	.01	.91
despicable	.01	.00	.01	.00	.00	.00	.94
malicious	.00	.00	.01	.02	.00	.02	.94
grotesque	.00	.02	.00	.00	.03	.05	.88
sleazy	.00	.01	.03	.02	.00	.02	.90
cruel	.04	.03	.03	.03	.00	.01	.88

Note. $n = 522$. The strongest loadings for each factor are bolded.

iors, important situations were also marked by extraverted, agreeable, and open behaviors. Thus, important situations might call for behaviors that enhance one's self-image or behaviors that can help to progress the situation toward a favorable conclusion through engaging in increasing amounts of communication, prosocial, and/or problem solving behaviors.

The hypothesized relations between the CAPTIONs and affect were only partially supported. The results reveal that, although positive affect was positively related with both Positive Valence and Humor, it was also positively related with Complexity and Importance (as well as with Adversity and Negative Valence, although the correlations were smaller). Although these findings were not hypothesized, they are conceptually interpretable: in complex and important situations, an individual might be more "alert," "active," "interested," and "attentive" (i.e., positive affect) because these situations are much more nuanced, intricate, and engaging. Similar to the findings for in situ personality-relevant behaviors, important and complex situations may call for increased attention to detail, enthusiasm/energy, and intellectual engagement. Negative affect was found to be positively related with both Negative Valence and Adversity, as expected. However, negative affect was also found to positively relate with Complexity and Humor. These relations may be indicative of the fact that complex situations are often higher in Adversity, which might cause increased negative affect. Similarly, situations which are juvenile/childish (Humor) might be more likely to cause annoyance and/or frustration.

The hypothesized relations of the CAPTION dimensions with need satisfaction and intrinsic motivation were partially supported. The predicted relations for both belonging and self-esteem were generally supported. However, both Adversity and Complexity were also found to be negatively related with belonging. As predicted, Negative Valence and Importance were found to have the strongest relations with meaningful existence. However, the hypothesized relations with Complexity did not emerge. It is quite possible that the pathway from these psychological situation dimensions to meaningful existence might be moderated by other factors, such as personality, in that they only lead to meaningful existence when certain conditions are met. For example, complex situations may only lead to meaningful existence when one is higher on openness to experience, as they might enjoy the cognitive and emotional complexity of the situation more so than others. Importance and Negative Valence, as predicted, found to be related with control, but the hypothesized association with Typicality did not emerge. Similar to the relation between Complexity and meaningful existence, it is possible that other factors (such as locus of control) might moderate this association. The hypothesized

relations of intrinsic motivation for Positive Valence and Humor were also found, whereas the predicted relations with Negative Valence and Importance were not. Additionally, although not hypothesized, Adversity showed moderate negative relations with intrinsic motivation, suggesting that frustrating situations may lead individuals to become less interested in the task at hand.

General and incremental validity of the CAPTIONs. We sought to assess whether the CAPTION taxonomy, as a whole, can capture meaningful variance in psychological outcomes, both in isolation as well as over and above two existing measures of situational characteristics—that is, van Heck's (1984) measure of situation classes and the personality-relevant contexts of the RSQ-8 (Rauthmann et al., 2014). The results of these analyses can be found in Table 11. As can be seen, the CAPTIONs collectively predicted a substantial amount of variance in each of the outcomes (i.e., personality-relevant behaviors, affect, need satisfaction, and intrinsic motivation) both in isolation ($R_s = .41 \sim .83, p < .05$) and beyond the two aforementioned existing measures of situations ($\Delta R^2_s = .04 \sim .14, p < .05$). The CAPTIONs was almost always a stronger predictor, both individually and incrementally, compared with the other two measures. In fact, the average incremental validity of the CAPTIONs ($\Delta R^2 = .09$) was roughly twice as large (1.73 and 2.21 times larger, respectively, using the recommendations made by Ozer, 2007 for calculating effect sizes) as either the RSQ-8 ($\Delta R^2 = .03$) or van Heck's dimensions ($\Delta R^2 = .02$), providing strong support for the breadth of the CAPTION taxonomy in capturing the broadest possible set of situation characteristics. These findings suggest that the breadth of the CAPTION model captures something beyond the existing situation measures, even within specific domains, such as predicting personality-relevant behaviors.

Relative importance of the CAPTIONs. In addition to testing whether the CAPTION model incrementally predicts the aforementioned outcomes above and beyond van Heck's measure and the RSQ-8, we also examined the relative importance of the CAPTION model in predicting these outcomes when considered together with the other two situational measures. This was done by running a series of relative weight analyses (Johnson, 2000; Toni-dandel & LeBreton, 2011, 2015; Tonidandel, LeBreton, & Johnson, 2009). As summarized in Figures 2 through 4, the results generally provide support for the CAPTION model's ability to predict important psychological outcomes.

For personality-relevant behaviors (see Figure 2), the results revealed that, despite its broader content coverage, the CAPTIONs was a relatively more important predictor of personality-relevant behaviors compared to both van Heck's measure of situation classes and the personality-relevant contexts of the RSQ-8, ac-

Table 10
Descriptives and Intercorrelations for Study Variables in Phase 4: Study 6

Variable	Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
<i>M</i>	2.45	2.17	2.76	3.70	3.47	2.22	1.49
<i>SD</i>	1.04	1.10	1.22	.85	.95	1.02	.87
CAPTIONs							
Complexity	(.94)						
Adversity	.51*	(.96)					
Positive Valence	.23*	-.02	(.97)				
Typicality	.04	.02	-.02	(.94)			
Importance	.43*	.27*	.33*	.21*	(.94)		
Humor	.30*	.25*	.61*	.07	.14*	(.96)	
Negative Valence	.43*	.48*	.30*	.08	.15*	.56*	(.98)
van Heck's dimensions							
Interpersonal conflict	.44*	.45*	.28*	.06	.21*	.48*	.60*
Joint working	.49*	.38*	.36*	.05	.39*	.35*	.31*
Intimacy	.36*	.26*	.39*	.05	.16*	.51*	.54*
Recreation	.14*	.00	.33*	-.03	-.05	.35*	.20*
Traveling	.28*	.32*	.29*	.04	.19*	.39*	.48*
Rituals	.40*	.24*	.41*	.07	.24*	.35*	.51*
Sport	.26*	.23*	.22*	-.06	.04	.34*	.44*
Excess	.35*	.26*	.32*	.04	.16*	.48*	.56*
Serving	.30*	.25*	.31*	.12*	.31*	.30*	.33*
Trading	.38*	.36*	.22*	.08	.21*	.32*	.46*
Edwards & Templeton items							
Positivity	.24*	-.06	.73*	-.13*	.24*	.51*	.28*
Negativity	.42*	.47*	.28*	.10*	.11*	.55*	.79*
Productivity/Goal achievement	.44*	.24*	.26*	.21*	.70*	.12*	.19*
Routineness/Ease of negotiation	-.08	-.41*	.47*	.26*	.02	.28*	.06
RSQ-8 (DIAMONDS)							
Duty	.48*	.51*	.04	.12*	.55*	.07	.26*
Intellect	.66*	.43*	.24*	.03	.43*	.24*	.36*
Adversity	.45*	.49*	.28*	.04	.19*	.50*	.70*
Mating	.34*	.26*	.46*	.05	.21*	.53*	.53*
Positivity	.17*	-.06	.58*	.03	.17*	.49*	.14*
Negativity	.48*	.70*	.04	.02	.22*	.30*	.46*
Deception	.47*	.53*	.21*	.05	.21*	.43*	.58*
Sociality	.31*	.24*	.43*	.03	.29*	.39*	.24*
Personality							
Honesty/Humility	-.30*	-.29*	-.20*	-.09*	-.21*	-.31*	-.37*
Emotionality	-.12*	.10*	.04	-.03	-.02	.06	-.03
Extraversion	.09*	-.08	.20*	-.03	.16*	.14*	.00
Agreeableness	-.08	-.16*	.11*	-.11*	.01	-.02	-.06
Conscientiousness	-.25*	-.30*	-.17*	-.02	-.03	-.31*	-.46*
Openness	.12*	-.05	-.01	-.03	.19*	-.06	-.15*
Personality behaviors							
Extraversion	.21*	.11*	.36*	-.04	.36*	.26*	.17*
Agreeableness	.09*	-.08	.35*	.01	.36*	.12*	-.05
Conscientiousness	.19*	.10*	.10*	.12*	.46*	-.06	-.03
Neuroticism	.10*	.34*	-.15*	-.08	-.13*	.03	.14*
Openness	.33*	.07	.27*	-.04	.31*	.14*	.11*
Affect							
Positive affect	.46*	.21*	.51*	-.03	.53*	.37*	.27*
Negative affect	.47*	.61*	.22*	.02	.13*	.50*	.77*
Need satisfaction							
Belonging	-.24*	-.42*	.13*	-.04	.15*	-.22*	-.51*
Self-esteem	.08	-.18*	.41*	.03	.39*	.13*	-.09*
Meaningful existence	-.11*	-.25*	.11*	-.03	.31*	-.20*	-.43*
Control	.09*	-.08	.17*	.08	.36*	.00	-.15*
Motivation							
Intrinsic motivation	.03	-.38*	.53*	-.16*	.06	.24*	-.08

Note. $n = 522$.

* $p < .05$.

Table 11
Incremental Validity of the CAPTIONs

Variable	In Isolation		Incremental	
	<i>R</i>	<i>R</i> ²	Step 1 ^a <i>R</i> ²	Step 2 ΔR^2
CAPTIONs				
Extraversion	.46	.21	.33	.05
Agreeableness	.48	.23	.30	.07
Conscientiousness	.48	.23	.25	.08
Neuroticism	.41	.17	.25	.04
Openness	.42	.18	.28	.04
Positive affect	.69	.47	.40	.14
Negative affect	.83	.69	.61	.13
Belonging	.65	.42	.42	.10
Self-esteem	.57	.33	.29	.12
Meaningful existence	.61	.37	.33	.13
Control	.44	.19	.26	.05
Intrinsic motivation	.69	.47	.48	.12
DIAMONDS				
Extraversion	.46	.21	.35	.03
Agreeableness	.47	.22	.34	.03
Conscientiousness	.42	.18	.31	.02
Neuroticism	.44	.19	.25	.04
Openness	.45	.20	.28	.04
Positive affect	.60	.36	.55	.03
Negative affect	.76	.58	.73	.02
Belonging	.62	.38	.54	.02
Self-esteem	.52	.27	.48	.02
Meaningful existence	.55	.30	.48	.03
Control	.47	.22	.29	.05
Intrinsic motivation	.66	.44	.57	.05
van Heck's				
Extraversion	.40	.16	.35	.03
Agreeableness	.35	.12	.36	.01 ^b
Conscientiousness	.28	.08	.32	.01 ^b
Neuroticism	.24	.06	.28	.01 ^b
Openness	.33	.11	.30	.02 ^b
Positive affect	.51	.26	.56	.02
Negative affect	.67	.45	.74	.01 ^b
Belonging	.50	.25	.53	.03
Self-esteem	.35	.12	.49	.01 ^b
Meaningful existence	.46	.21	.49	.02 ^b
Control	.36	.13	.32	.02 ^b
Intrinsic motivation	.48	.23	.59	.03

^a In Step 1, the two other situation measures were entered. For personality-relevant behaviors, the HEXACO personality factors were also entered. All *R* and ΔR^2 values were statistically significant at $p < .05$ with the exception of those indicated by ^b.

counting for approximately 38% to 70% of the total predicted variance in the criteria. The only exception to this was for the neurotic behaviors, where the largest proportion (55.92%) of the total predicted variance was attributed to the RSQ-8, followed by 37.51% being attributed to the CAPTIONs. This may be better understood in light of our earlier discussion that the DIAMONDS model includes multiple Negative Valence-related facets. As facets often provide richer (and more fine-grained) descriptions of the content domain compared to general factors (e.g., Judge, Rodell, Klinger, Simon, & Crawford, 2013; Paunonen & Ashton, 2001) and "negative" is often stronger than "positive" (e.g., Baumeister et al., 2001), we would expect that these dimensions capturing more nuanced variations in the Negative Valence of situations might collectively contribute more to the prediction of neurotic behaviors than the broad Negative Valence dimension of the CAPTIONs, in isolation. Overall, the analyses revealed that the

CAPTIONs, on average, predicted more variance in personality-relevant behaviors (49.50%) than either of the extant measures (RSQ-8 = 41.26%; van Heck = 9.24%), despite the fact that the CAPTION model was not directly derived in order to assess situations in which personality-relevant behaviors might manifest and in spite of the breadth of the CAPTION dimensions.

In addition to the strong predictive validity found for personality-relevant behaviors, the CAPTIONs was also found to generally account for larger proportions of the predicted variance in the other outcomes (see Figures 3 and 4). Just as for personality-relevant behaviors, the CAPTIONs was found to be the most important predictor of affect, predicting approximately 56% and 41% of positive and negative affect, respectively (RSQ-8 = 31.46% & 31.55%; van Heck = 12.10% & 27.70%). Similar results were found for intrinsic motivation, with the CAPTIONs being the more important predictors of the outcome (43.69%) than either the RSQ-8 dimensions (38.89%) or the dimensions of van Heck (17.42%). This trend largely remained when examining need satisfaction. Overall, the CAPTIONs predicted, on average, more relative variance (48.21%) than either the other two measures (RSQ-8 = 38.32%; van Heck = 13.48%). The only exception to this pattern of findings was that the RSQ-8 was a larger relative predictor of control (50.95%), than the CAPTIONs (36.22%) or van Heck's (12.84%). Overall, these findings provide strong support for the predictive validity, breadth, and integrative nature of the CAPTION dimensions, even in the prediction of more specific outcomes.

Creating the Short-Form CAPTIONs. Although the 70-item CAPTIONs provides a useful and important tool to assess the major dimensions of situations, practical constraints of research often necessitate a shorter measure. Mindful of this, we sought to develop a shortened version of the CAPTIONs, which we refer to as the CAPTIONs Short-Form or CAPTIONs-SF. To derive the CAPTIONs-SF, we used Ant Colony Optimization (ACO; Leite, Huang, & Marcoulides, 2008; Marcoulides & Drezner, 2003), which was developed as a methodology for addressing complex, combinatorial optimization problems based upon the foraging behaviors of ants. When foraging, ants initiate their food-gathering attempt by randomly advancing toward a source of food, with each ant leaving behind a trail of pheromones. As these initial random permutations continue, the pheromone trails across the shortest pathways become stronger due to more frequent travel. The ants, drawn to the pheromones, increasingly begin to follow these stronger-pheromone pathways until the optimal foraging pathway is revealed. Similarly, the ACO algorithm begins with random short-form permutations of the full measurement model (i.e., all items are equally sampled from), and iteratively increases the sampling weights for those items which produce a better model fit, eventually converging upon an optimal short-form solution. The ACO technique has been shown to outperform other short-form methodologies, such as maximizing reliability/primary loadings, minimizing fit indices/cross loadings, and other optimization algorithms (Olaru, Withöft, & Wilhelm, 2015). We implemented the ACO algorithm, seeking to derive a short-form measure with four items per factor (28 items, in total). To find the optimal short-form solution, we ran the ACO algorithm 20 times, varying the number of ants and iterations used (for a total of 49,500 individual runs), and retained the items which were the most consistently found. Ties were broken based upon conceptually

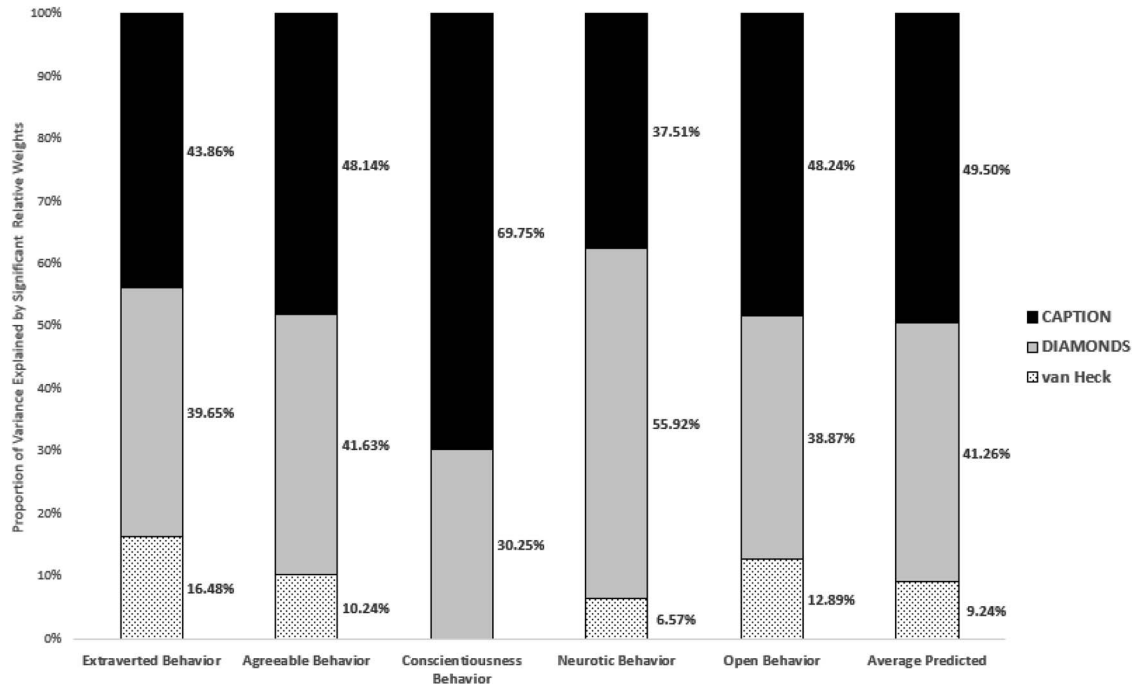


Figure 2. Relative importance of the CAPTIONs in predicting personality-relevant behaviors. The final column represents the average relative importance across the five personality-relevant behaviors.

optimizing the breadth of content coverage of the final short-form scale. The items of the CAPTIONs-SF are noted by asterisks in Appendix C.

Detailed findings of the validity analyses can be found in the supplemental materials (Tables 2–4; Figures 1–3). The ESEM results revealed that the CAPTIONs-SF fit the data very well ($\chi^2 = 343.76$, $df = 203$, $CFI = .99$, $TLI = .98$, $RMSEA = .04$, and $SRMR = .01$), with strong primary loadings and negligible cross-loadings. The reliabilities for the CAPTIONs-SF scales were also strong ($\alpha s = .86$ – $.97$). Most importantly, the CAPTIONs-SF showed impressive convergent validity with the relevant full-scale CAPTIONs dimensions ($r s = .95$ – $.98$) and maintains the same pattern of relations with the outcomes of interest, supporting the effectiveness of the CAPTIONs-SF in assessing the broad CAPTION dimensions. We also reran the incremental validity and relative importance analyses conducted for the full-scale CAPTIONs and found that the CAPTIONs-SF had nearly identical incremental and relative predictive validity to that of the CAPTIONs full-form scale. The CAPTIONs-SF, despite having fewer dimensions than either of the extant measures, predicted unique variance above and beyond these measures ($\Delta R^2 s = .03$ – $.14$, $p < .05$). Additionally, its average incremental validity ($\Delta R^2 = .09$) remained approximately twice that of either the RSQ-8 ($\Delta R^2 = .03$) or van Heck's dimensions ($\Delta R^2 = .02$), using recommendations for effect size calculation made by Ozer (2007), just as was the case with the full-form of the CAPTIONs. The relative importance analyses also found a similar pattern of findings to that of the full-form scale, with the CAPTIONs-SF generally predicting more relative variance across the outcomes. Again, despite the fact that the CAPTION dimensions were not directly derived in order to predict personality-relevant behaviors, the CAPTIONs-SF, with fewer di-

mensions, predicted more relative variance, on average, across these outcomes (50.31%) than either of the extant measures (RSQ-8 = 38.99%; van Heck's = 10.70%). These results provide strong evidence that the CAPTIONs-SF offers a parsimonious way to assess the broad content domain of psychological situation characteristics.

Summary. Taken together, these findings provide evidence for the validity of the CAPTION model. We found strong evidence that the hypothesized seven-dimensional structure can be effectively assessed by both the 70-item scale (CAPTIONs) and the 28-item scale (CAPTIONs-SF). Additionally, the CAPTION model was found the strongest predictor of important psychological outcomes, both *in addition* and *relative* to other measures of psychological situation characteristics in both its full- and short-form measurement structure. The results support the notion that a lexically derived taxonomy captures the broadest sets of descriptors available and can integrate the findings from taxonomies derived using differing conceptual lenses to predict important outcomes (e.g., John & Srivastava, 1999; Ten Berge & De Raad, 1999; Yang et al., 2006). Overall, these results provide strong support for the CAPTIONs and the CAPTIONs-SF in assessing the full, broad content domain of psychological situation characteristics.

General Discussion

The current paper presents the largest-to-date lexical analysis of psychological situation characteristics. This analysis led to the identification and replication of seven major dimensions of situations emerging through a diverse array of derivation methodologies including qualitative inputs, large-scale in situ ratings of

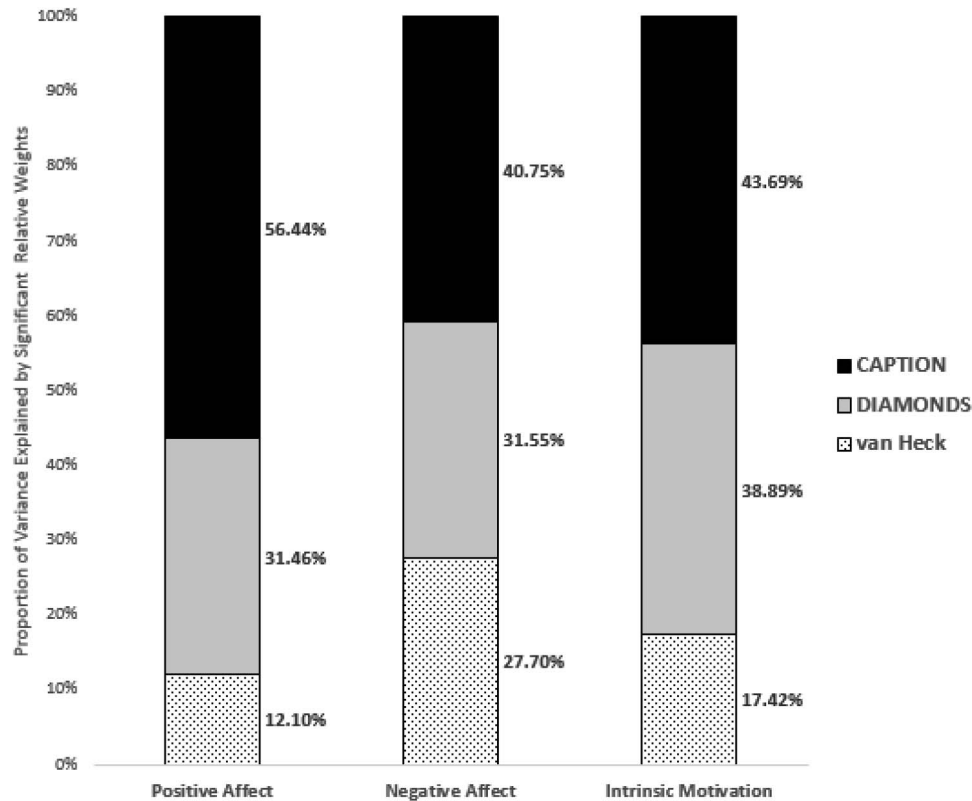


Figure 3. Relative importance of the CAPTIONs in predicting motivation and affect.

situations using differing item content, and natural-language use via lexical vector representations from neural-network-based models. As these seven dimensions were derived via a lexical approach, they do not place the same, narrower, a priori restrictions on the content domain of situation characteristics being sampled as non-lexical taxonomic approaches. Thus, the current research represents a meaningful step toward overcoming what has been one of the most consistently bemoaned issues in situational taxonomic efforts: the simultaneous analysis of a more fully representative collection of situation characteristics. The seven-dimensional structure that emerged from our investigations entails Complexity, Adversity, Positive Valence, Typicality, Importance, Humour, and Negative Valence, which we collectively refer to as the CAPTION model of situation characteristics.

The current article also introduces a 70-item measure of these seven broad CAPTION dimensions (the CAPTIONs), and a short-form 28-item measure (the CAPTIONs-SF), along with some compelling initial validity evidence, which are intended to serve as useful tools for advancing our understanding of how psychological situations may influence important psychological outcomes. Evidence was found for both the incremental and the relative predictive validity of the CAPTION model, supporting the ability of the lexical hypothesis to integrate findings using differing conceptual lenses as well as the validity of the CAPTION dimensions in predicting important psychological outcomes. The results also suggest that an eventual facet-level exploration of the CAPTION dimensions may be useful from both a theoretical and a practical standpoint. Overall, in the current study we found a stable, repli-

cable, and broad dimensional structure of psychological situation characteristics; developed and validated psychometrically sound measures for these dimensions; and demonstrated that these dimensions can predict substantial variance in important psychological outcomes.

A final point worthy of discussion is the evaluative nature of psychological situation characteristics. The CAPTION taxonomy does not seek to exclude evaluative terms, but rather actively incorporates them. One could argue that situations should best be described in evaluation-free language, as this would be more “objective.” Indeed, if one simply wishes to identify the objective description of the situation (i.e., who, what, when, where, which), then situation assessment via psychological situation characteristics should not be used. However, the very essence of the psychological situation is the psychologically salient, meaning-generated evaluations of the objective situation (Block & Block, 1981; Cantor et al., 1982; Ekehammar, 1974; Endler & Magnusson, 1976; Lewin, 1936, 1951; Murray, 1938; Rauthmann, Sherman, & Funder, 2015; Shoda et al., 1994)—assessed at either the individual or the consensual/canonical level (Block & Block, 1981). Similarly, although attempts have been made to remove evaluation from the personality sphere, it is quite clear that “Personality attributes do not exist in language or in psychologists’ inventories merely for decontextualized and bloodless description . . . personality terms are evaluative rather than neutral” (Saucier & Srivastava, 2015, p. 295). This same principle applies to psychological situation research—individuals do not process situations as some objective, omnipotent being. Rather, individuals are actively per-

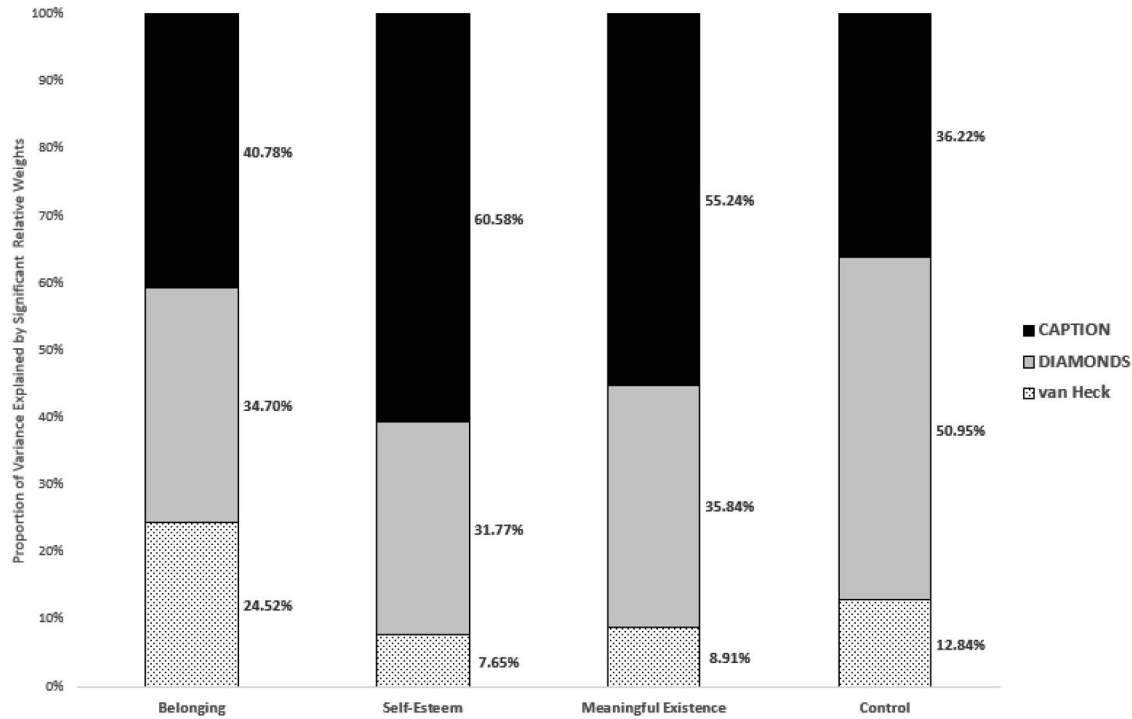


Figure 4. Relative importance of the CAPTIONs in predicting need satisfaction.

ceiving and processing the situations they are in using their current set of schemas, goals, experiences, and other biased mechanisms. Thus, it is quite clear that dimensions such as Positive Valence, Negative Valence, Adversity, and so forth, which have continually been found in the literature, would remain evaluative in nature, in spite of efforts to the contrary. In sum, the removal of evaluative content of the psychological situation is simply not possible, much less desired. Rather, this evaluative content is actually the essence of the psychological situation, and is what distinguishes it from its objective situation counterpart.

Limitations and Implications for Future Research

We suggest that the major goals of future situational research should generally be toward the following: (a) clarifying the specific roles that situations play in individuals' lives; (b) increasing the understanding of person and situation interactions; (c) identifying the lower-order factor structure of the CAPTION model; (d) replicating the CAPTION dimensions across languages; (e) further clarifying the theoretical implications of these descriptive situation dimensions; and (f) the continued exploration of the adequacy of adjective-based taxonomic efforts. We elaborate on each point in the following paragraphs.

First, the question of *how* situations influence important psychological outcomes is integral to advancing the field and should be further explored. The wealth of theory and empirical research suggests that the psychological situation is one of the fundamental mediating mechanisms between the objective situation and psychological outcomes (e.g., Cantor et al., 1982; Edwards & Templeton, 2005; Ekehammar, 1974; Mischel & Shoda, 1995; Ross & Nisbett, 1991). In light of this, we envision that a vast amount of

empirical research can be conducted across multiple disciplines to uncover the nuanced relations between situational characteristics and psychological outcomes using CAPTION as an organizing framework. For example, objective features of situations can be mapped onto their corresponding psychological characteristics (e.g., how individuals generally perceive "running into a childhood friend in a business negotiation meeting" or "traveling on New Year's Day") to provide more direct insights into how situations may trigger certain affective, cognitive, and behavioral reactions in individuals. Also, using the CAPTION dimensions, studies may be able to look into the long-term effects of being exposed to certain psychological situations on individuals' life outcomes (e.g., how does perceiving one's daily situations as "important and positive" as opposed to "typical and negative" lead to his or her well-being and performance) in a more systematic, standardized manner. Lastly, psychological situation characteristics can be incorporated into experimental designs to ensure that the manipulations are effective, to more accurately manipulate situations, and to better understand the mechanisms through which the manipulations to the objective situation leads to psychological outcomes.

Second, as both person and situation variables are crucial for understanding psychological outcomes, the interactions between persons and situations needs to be further explicated. In the current article we provided conceptual and empirical evidence suggesting that the CAPTION dimensions capture a wide array of psychological situations in a way that is distinct from other key predictors of psychological outcomes such as personality. We have also established initial support for predictive relations between the CAPTION dimensions and important psychological outcomes (i.e., personality-relevant behaviors, affect, need satisfaction, and

intrinsic motivation). Building from this work, several key questions can be examined to further clarify the dynamic interplay between personality and situations: How do certain personality characteristics (e.g., conscientiousness) change the fundamental way one perceives, and subsequently acts, in a particular situation (e.g., an important situation)?; Do individuals with higher levels of cognitive ability, self-efficacy, or experience within a particular domain perceive situations in more effective ways?; Which ways do individuals with depression and/or anxiety differentially perceive the world from individuals who do not have such a diagnosis? All of these questions, and others, can provide interesting and important insights for many areas of psychology and beyond. Although we have provided preliminary data on how individuals with different personality traits might systematically perceive and/or encounter particular types of situations with greater frequency (e.g., highly agreeable individuals may be less likely to perceive and/or encounter situations with high levels of Negative Valence), we are unable to untangle the directionality and dynamics of such relations. However, it is our hope that this situational taxonomy will provide an empirical basis from which researchers can further explore these complex relations.

Third, from a measurement perspective, although the broad CAPTION dimensions have consistently emerged across diverse methods, an examination of their lower-order structure is also needed. As discussed previously, the number and richness of the situation characteristics made available by using the lexical hypothesis would allow us to uncover the facets underlying the broad dimensions of the psychological situation. Similar to how the lexical hypothesis has allowed for a multitude of fine-grained explorations of the lower-level structure of personality (e.g., Costa, McCrae, & Dye, 1991; DeYoung et al., 2007; Goldberg, 1999; Lee & Ashton, 2010), the broad, comprehensive nature of the lexically derived psychological situation characteristics should allow for similar understandings for the lower-level structure of major situation dimensions.

Fourth, even though the current study used a variety of methodologies and analytic techniques to identify the consistent lexically derived dimensions of situations, conclusions from this study are limited entirely to the English language. To determine whether the CAPTION dimensions are universal across languages and cultures, cross-language/cultural replications are required. It is quite possible that slight differences across languages might emerge. The goal for this taxonomic structure is parallel to that of personality, where the core dimensions of personality have consistently been replicated across many languages and cultures (e.g., De Raad, Perugini, Hrebícková, & Szarota, 1998; McCrae & Costa, 1997). It is our hope that findings from our study will provide useful guidelines for the continued research efforts toward replicating and refining the major dimensions of situations.

Fifth, although the lexical hypothesis as applied to situations is theoretically grounded within the evolution of language (e.g., Corballis, 2009; Kirby, 2007; Pinker & Bloom, 1990), the taxonomy remains descriptive in nature. We believe that the first step in identifying the broad construct space is to first identify the *what*, and then to move onto the *why*. So, just as the Big Five or the HEXACO models of personality are largely derived via atheoretical, bottom-up processes (e.g., Ashton & Lee, 2005; Hampson, 2012), the CAPTION model is also first and foremost a descriptive taxonomy of the psychological situation. It is our hope that rich

theoretical explanations of this descriptive structure, similar to those emerging within the personality sphere (e.g., Fleeson & Jayawickreme, 2015), can proliferate now that an integrative situational dimensional structure is being identified. Although a wealth of potential avenues for this theoretical structure exists, we believe that, consistent with other researchers (e.g., Brown, Neel, & Sherman, 2015; Rauthmann, 2015; Yang et al., 2006, 2009) the theoretical integration of goals and situations likely provides one of the most efficient means of beginning this theoretical progression toward explanation.

Last, similar to concerns raised regarding the adequacy of using single-term descriptors to assess personality (e.g., Ashton & Lee, 2005), concerns regarding how best to assess situations have also been discussed (Saucier, Bel-Bahar, & Fernandez, 2007). Specifically, researchers have suggested that situations might need to be described by full sentences, such as “my doctor gave me bad news in his office,” to adequately assess the nuances involved within the situation. It is true that abstract adjectives are inadequate in describing the detailed content of who, what, when, where, and which (i.e., the highly detailed objective situation). Although situations do contain a host of nuanced information, the goal of a situational taxonomy is not to capture every nuance within situations. Rather, the goal of a situational taxonomy should be to be able to integrate this highly detailed psychological content into its formal, abstract properties which are the “generalizations and uniformities spanning situations” (Block & Block, 1981, p. 90). In other words, the goal is to identify the characteristics of situations which span situations, rather than identifying the rich minutia present within a specific situation. Practically, adequately capturing the seemingly endless possible full sentences describing situations would be an extremely challenging, if not impossible, task. Thus, although there exists a rich amount of detail at the classificatory level of psychological content (i.e., the objective situation), the goal of a situational taxonomy is to integrate these specific details within a larger, more parsimonious framework (Block & Block, 1981), such as through single-term descriptors which succinctly integrate these innumerable objective nuances.

Similar to the lexical derivations of personality, the CAPTION taxonomy does not seek to fully capture the entirety of the objective situation. Instead, the goal is to provide a framework which captures the major dimensions thereof. It is also worth noting that the lexically derived CAPTION model fully integrates the extant dimensions found in the literature from studies employing a variety of methodological and conceptual approaches. This also supports the notion that adjectives are an effective method for assessing the formal, abstract properties of situations. Although adjectives are likely the most parsimonious method for capturing the broad domain of psychological situation characteristics, this remains an empirical question that is open to scrutiny. Thus, we believe that explorations using more nuanced descriptions of situations (i.e., full sentences) could prove quite useful.

Conclusion

By leveraging the strength of the lexical hypothesis in assessing the full content space of psychological situations and integrating research using more specific conceptual lenses, we have uncovered seven major dimensions of psychological situation characteristics that have consistently emerged across a diverse array of method-

ologies. While identifying a taxonomic structure may only be the beginning of a long journey toward the systematic understanding of situations in psychological science, we hope that this taxonomy and the corresponding measurement tools we developed and validated in this article will help to make that path a well-trodden one.

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(Appendices follow)

Appendix A

List of 851 Psychological Situation Characteristics

abnormal	abrasive	absurd	academic	active
adequate	admirable	affectionate	aggressive	agitating
agonizing	agreeable	alarming	alien	alluring
amazing	ambitious	amusing	analytical	annoying
appalling	appropriate	argumentative	artificial	astonishing
average	awful	backhanded	bad	baffling
barbaric	bearable	beneficial	benign	bewildering
bittersweet	bizarre	bland	bleak	blissful
boring	breathtaking	brilliant	brisk	broadminded
brutal	burdensome	callous	calming	candid
captivating	carefree	careful	careless	casual
catastrophic	ceremonial	challenging	changeable	chaotic
charming	cheerful	cheerless	cheesy	cherished
childish	chilling	civil	classless	classy
claustrophobic	clear	clever	coherent	collaborative
colorful	combative	comfortable	comforting	comfortless
comical	commendable	common	compassionate	compelling
competitive	complex	complicated	comprehensible	conceivable
condescending	confrontational	confusing	consequential	consistent
constrictive	contemplative	controllable	controversial	convenient
conventional	cooperative	cordial	corruptive	costly
counterproductive	courteous	covert	cozy	crappy
crazy	creative	creepy	critical	crucial
crude	cruel	crummy	cultured	curious
cute	dangerous	daring	dark	daunting
deadly	debilitating	decent	deceptive	deep
delicate	delightful	deplorable	depressing	derogatory
desirable	desperate	despicable	destructive	detestable
devastating	developmental	devilish	devious	different
difficult	dignified	diplomatic	dirty	disagreeable
disappointing	disastrous	discomforting	discouraging	discreet
disgraceful	disgusting	dishonest	dismal	disorderly
disrespectful	distinctive	distressing	divine	dizzying
dreadful	dreary	dull	dumb	dysfunctional
easy	edgy	educational	eerie	effective
efficient	effortless	elaborate	elementary	embarrassing
emotional	emotionless	enchanted	encouraging	endearing
endurable	energetic	enjoyable	enlightening	entertaining
enticing	entrepreneurial	epic	erotic	erratic
eternal	ethical	eventful	exasperating	excellent
exceptional	excessive	exciting	excusable	exhausting
exhilarating	exotic	exploitative	explosive	exquisite
extensive	extraordinary	extravagant	extreme	fabulous
fair	fake	familiar	fancy	fantastic
fascinating	fatiguing	fierce	filthy	fine
finicky	fishy	flattering	flawed	flexible
flirtatious	fluctuating	foolish	foregone	foreign
forgettable	forgiving	formal	formative	foul
frantic	freaky	fretful	friendly	frightening
frightful	fruitful	frustrating	fun	functional
funky	funny	geeky	glamorous	gleeful
gloomy	glorious	gluttonous	good	goofy
graceless	gratifying	grave	great	grim
groovy	gross	grotesque	grueling	gutsy
habitual	hairy	halting	handy	happy
hard	harmful	harmless	harmonious	harsh
hasty	hateful	hazardous	heartbreaking	heartfelt
heartless	heartwarming	heavenly	hectic	hellish
helpful	helpless	hilarious	honorable	hopeful

(Appendices continue)

Appendix A (continued)

abnormal	abrasive	absurd	academic	active
hopeless	horrendous	horrifying	hospitable	hostile
humbling	humiliating	humorless	humorous	hurtful
hyperactive	hypnotic	hysterical	icky	ideal
idealistic	idiotic	illogical	illuminating	immature
immoral	impactful	impassioned	impersonal	implausible
impolite	important	impossible	improbable	improper
impure	inadvisable	inappropriate	incomprehensible	inconceivable
inconsequential	inconvenient	incredible	incriminating	indecent
indefinable	indescribable	industrious	inebriating	ineffectual
inevitable	inexcusable	inexplicable	inflammatory	inflexible
informal	informational	infuriating	inhospitable	inhumane
innocent	inoffensive	insane	insensitive	insignificant
insincere	inspirational	instructional	instrumental	insufferable
insulting	intellectual	intelligent	intelligible	intense
interesting	intimate	intimidating	intolerable	intricate
intriguing	intuitive	invaluable	inventive	invigorating
ironic	irrational	irregular	irresistible	irresponsible
irritating	joyful	joyless	jubilant	judgmental
juvenile	kinky	lame	legendary	lengthy
lethal	liberating	lifeless	lifesaving	lighthearted
lively	lonely	lonesome	longstanding	lousy
loveless	lovely	loving	luckless	lucky
lucrative	ludicrous	magical	magnificent	majestic
malicious	manageable	marvelous	meaningful	meaningless
mediocre	meditative	mellow	melodramatic	memorable
menacing	merry	messy	mighty	mindless
minor	miraculous	mischievous	miserable	misleading
momentous	monstrous	morbid	motivational	multidimensional
multilayered	multipurpose	murky	mystical	nasty
natural	naughty	nauseating	navigable	needless
negative	neglectful	nerdy	neutral	new
nice	nightmarish	noble	noncompetitive	nonfunctional
nonsexual	nonviolent	normal	notable	nourishing
nutty	objectionable	obligatory	obnoxious	obscure
observable	obvious	odd	offbeat	offensive
oppressive	optimistic	orderly	ordinary	organic
outrageous	outstanding	overambitious	overblown	overdramatic
overemotional	overrated	overwhelming	paralyzing	passionate
passionless	passive	pathetic	peaceful	peculiar
perfect	permissive	perplexing	personal	perverted
petty	phenomenal	phony	pitiful	playful
pleasant	pleasurable	pointless	polite	political
positive	powerful	practical	praiseworthy	precious
predictable	prejudicial	preposterous	prestigious	priceless
private	proactive	problematic	productive	professional
profound	prohibitive	promising	proper	prosperous
provocative	prudent	psychotic	purposeless	puzzling
queer	questionable	quick	quirky	racy
radical	random	rapid	rare	rational
reasonable	reckless	refined	refreshing	regrettable
regular	relaxing	relentless	remarkable	repetitious
reprehensible	repulsive	reputable	respectful	restful
restless	restorative	restraining	restrictive	revitalizing
revolting	rewarding	ridiculous	rigid	risky
ritualized	riveting	romantic	rotten	rough
rowdy	rude	ruthless	sacred	sad
sadistic	sassy	satisfying	scandalous	scary
scholarly	scientific	scrappy	screwy	seductive
selfish	selfless	semiformal	sensational	senseless
sensible	sensitive	sensual	sentimental	serene
serious	sexual	shady	shaky	shallow

(Appendices continue)

Appendix A (continued)

abnormal	abrasive	absurd	academic	active
shameful	shocking	shortsighted	significant	silly
simple	simplistic	sincere	sinful	sinister
sinless	sketchy	sleazy	slow	sobering
solemn	somber	soothing	sophisticated	soulful
special	spectacular	spirited	spiritual	splendid
spontaneous	spooky	stabilizing	stable	stagnant
stale	standard	startling	static	steamy
stiff	stimulating	stinky	straightforward	strange
strategic	strenuous	stressful	stuffy	stunning
stunted	stupid	subdued	substantial	subtle
successful	super	superb	superficial	supernatural
supportive	surprising	surreal	survivable	suspenseful
suspicious	sympathetic	tacky	tactful	tactical
tantalizing	tasteful	technical	tedious	temperamental
tender	tense	terrible	terrific	terrifying
thankless	theatrical	therapeutic	threatening	thrilling
tiresome	tiring	tolerable	torturous	touchy
tough	toxic	tragic	tranquil	transformative
transparent	trashy	traumatic	treacherous	treasured
tricky	triumphant	turbulent	typical	unaccommodating
unadvised	unattainable	unauthorized	unavoidable	unbearable
unbelievable	uncaring	uncertain	unchecked	uncivilized
unclean	unclear	uncomfortable	uncommon	uncommunicative
uncomplicated	unconquerable	uncontestable	uncontrollable	uncontroversial
unconventional	uncritical	uncultured	undemanding	underhanded
understandable	undesired	undignified	undirected	uneasy
unemotional	unending	unendurable	unenlightened	unenlightening
unerotic	unessential	unethical	uneventful	unexceptional
unexciting	unexpected	unexplainable	unfair	unfamiliar
unfavorable	unforeseeable	unforeseen	unforgettable	unforgivable
unforgiving	unfortunate	unfriendly	ungodly	ungraceful
unhappy	unhelpful	unifying	unimpressive	uninhibited
uninspired	unintelligent	uninteresting	uninvolved	unique
unjust	unjustified	unlovely	unloving	unlucky
unmerciful	unnatural	unneeded	unorganized	unorthodox
unpleasant	unpleasing	unpredictable	unprincipled	unproductive
unprofessional	unpromising	unreasonable	unrecognizable	unrefined
unregulated	unrehearsed	unrelenting	unremarkable	unrestrained
unrestricted	unrevealing	unrewarding	unromantic	unsafe
unsatisfying	unsavory	unscientific	unscripted	unsentimental
unsolvable	unsophisticated	unsportsmanlike	unstable	unsteady
unstructured	unsubstantial	unsuccessful	unsympathetic	unusual
unwanted	unwelcome	unwholesome	unwise	unyielding
upbeat	uplifting	upright	useful	useless
usual	vague	validating	valuable	vibrant
vicious	vile	vindictive	violent	virtuous
vital	vivacious	volatile	vulgar	wacky
warlike	warm	warped	weak	weird
welcoming	whimsical	wicked	wild	witty
wonderful	worrisome	worthless	worthwhile	yucky
zestful				

(Appendices continue)

Appendix B

List of 535 Psychological Situation Characteristics

active	adequate	affectionate	aggressive	agonizing
alarming	amazing	ambitious	amusing	analytical
annoying	appalling	appropriate	argumentative	artificial
astonishing	awful	bad	barbaric	bearable
beneficial	bewildering	bittersweet	blissful	boring
brehtaking	brilliant	broadminded	brutal	burdensome
captivating	carefree	careful	careless	casual
ceremonial	challenging	changeable	chaotic	cheesy
cherished	childish	chilling	classy	claustrophobic
clear	clever	coherent	colorful	comfortable
comforting	comfortless	compassionate	compelling	competitive
complex	complicated	comprehensible	conceivable	condescending
confusing	controllable	controversial	convenient	cooperative
corruptive	costly	counterproductive	covert	crazy
creative	creepy	critical	crucial	cruel
curious	cute	dangerous	daring	dark
deadly	decent	deceptive	delightful	deplorable
depressing	derogatory	desirable	desperate	destructive
devastating	developmental	devious	different	difficult
diplomatic	disagreeable	disappointing	disastrous	discomforting
discouraging	discreet	disgraceful	disgusting	dishonest
disorderly	disrespectful	divine	dizzying	dreadful
dysfunctional	easy	edgy	educational	effective
efficient	effortless	elaborate	elementary	embarrassing
emotional	endearing	enlightening	entertaining	entrepreneurial
epic	ethical	eventful	excessive	exciting
excusable	exhausting	exotic	exploitative	explosive
exquisite	extensive	extraordinary	extreme	fair
fake	familiar	fancy	fierce	finicky
flattering	flawed	flexible	flirtatious	fluctuating
foolish	foreign	forgettable	forgiving	formal
formative	foul	freaky	fretful	friendly
frustrating	fun	functional	funky	funny
gloomy	gluttonous	good	grave	great
grotesque	happy	harmful	harmless	harmonious
hasty	hateful	heartless	heartwarming	hectic
helpful	helpless	hilarious	honorable	hopeful
hopeless	horrifying	hostile	humbling	humiliating
humorless	hyperactive	hypnotic	icky	idealistic
immoral	impactful	impersonal	implausible	important
impossible	improbable	improper	impure	inadvisable
inconvenient	incredible	incriminating	indecent	indescribable
industrious	inebriating	inevitable	inflammatory	infuriating
inhospitable	inhumane	innocent	inoffensive	insensitive
insignificant	insincere	inspirational	instrumental	insulting
intellectual	intelligent	intense	interesting	intimidating
intriguing	intuitive	invaluable	invigorating	ironic
irrational	irresistible	irresponsible	joyful	judgmental
kinky	lame	lengthy	liberating	lifeless
lifesaving	lighthearted	lively	lonely	lousy
lovely	lucky	lucrative	magical	malicious
marvelous	meaningful	meditative	mellow	melodramatic
memorable	messy	mindless	minor	miraculous
mischievous	miserable	misleading	momentous	morbid
motivational	multidimensional	multipurpose	mystical	nasty
natural	navigable	needless	negative	nerdy
neutral	new	nice	nightmarish	noncompetitive
nonfunctional	nonsexual	nonviolent	normal	nourishing
obligatory	obnoxious	obscure	observable	obvious

(Appendices continue)

Appendix B (continued)

active	adequate	affectionate	aggressive	agonizing
oppressive	orderly	organic	outrageous	overambitious
overblown	overenthusiastic	overrated	overwhelming	paralyzing
passionate	passionless	passive	pathetic	perfect
permissive	personal	perverted	petty	playful
pleasant	pointless	polite	political	powerful
practical	praiseworthy	predictable	prejudicial	prestigious
priceless	proactive	problematic	productive	professional
profound	prosperous	provocative	psychotic	puzzling
questionable	quick	quirky	random	rare
reasonable	refined	refreshing	regrettable	regular
relaxing	relentless	remarkable	repetitious	reputable
respectful	restful	restless	restorative	restrictive
ridiculous	rigid	riveting	romantic	rowdy
rude	ruthless	sacred	sad	sassy
satisfying	scandalous	scary	scholarly	scientific
scrappy	screwed	seductive	selfish	selfless
semiformal	senseless	sensitive	sentimental	serious
sexual	shameful	shameless	shocking	shortsighted
silly	simple	simplistic	sincere	sinless
sleazy	slow	sobering	somber	soulful
special	spiritual	spontaneous	spooky	stabilizing
stable	stagnant	stale	steamy	stimulating
straightforward	strange	strategic	strenuous	stressful
stuffy	stunted	stupid	substantial	subtle
successful	superb	superficial	supernatural	supportive
surprising	surreal	survivable	suspenseful	suspicious
tacky	tasteful	technical	temperamental	tense
thankless	therapeutic	threatening	tiring	torturous
tough	toxic	tragic	transformative	transparent
traumatic	tremendous	tricky	triumphant	turbulent
typical	unadvised	unattainable	unauthorized	unbearable
uncaring	uncertain	unclear	uncomfortable	uncommon
uncommunicative	unconquerable	uncontestable	uncontrollable	uncontroversial
unconventional	uncritical	undignified	unemotional	unending
unenlightening	unerotic	unessential	unethical	uneventful
unexplainable	unfair	unforgivable	unforgiving	unfriendly
ungodly	ungraceful	unhelpful	unifying	uninhibited
uninspired	unintelligent	uninvolved	unjustified	unlovely
unloving	unlucky	unpleasant	unpredictable	unprincipled
unproductive	unprofessional	unpromising	unreasonable	unrecognizable
unrefined	unregulated	unrehearsed	unremarkable	unrestricted
unrevealing	unrewarding	unromantic	unsafe	unsatisfying
unsavory	unscientific	unsentimental	unsolvable	unsophisticated
unsportsmanlike	unstable	unstructured	unsuccessful	unsympathetic
unwanted	unwholesome	unyielding	upbeat	useless
validating	valuable	vindictive	violent	virtuous
vivacious	volatile	vulgar	warlike	warped
weak	wicked	wild		
worthwhile	zestful			

(Appendices continue)

Appendix C
Final Items of the CAPTIONs

Complexity	Adversity	Positive Valence	Typicality	Importance	Humor	Negative Valence
analytical*	stressful*	heartwarming*	typical*	effective*	wacky*	repulsive*
academic*	fatiguing*	cherished*	regular*	useful*	mischievous*	despicable*
scholarly*	frustrating*	precious*	standard*	productive*	goofy*	malicious*
instructional*	tiresome*	sentimental*	usual*	helpful*	childish*	grotesque*
complex	exhausting	loving	predictable	crucial	funny	vile
technical	tiring	affectionate	common	important	silly	inhumane
intricate	difficult	joyful	average	valuable	nutty	sinister
intellectual	strenuous	special	normal	vital	crazy	creepy
scientific	grueling	heavenly	ordinary	beneficial	juvenile	sleazy
educational	hectic	magnificent	uneventful	functional	immature	cruel

* Items of the 28-item CAPTIONs-SF.

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